Options Data and Models of Crude Oil Market with Idiosyncratic and Jump Risks*

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November 2, 2023

Abstract

I study the oil options market and observe two salient features: (i) deep out-of-the-money calls exhibit significantly negative average excess returns, and (ii) the $\text{VIX}_t^{\text{oil}}$ is high and consistently exceeds $\text{VIX}_t^{\text{S\&P500}}$. To explain these observations, I propose a model with idiosyncratic, jump, unspanned, and spanned risks. I derive expressions for futures prices, option prices, and option risk premiums, and use Kalman filtering and maximum-likelihood to estimate the model. My results show that the model simultaneously explains the diverse behaviors of option risk premiums, $\text{VIX}_t^{\text{oil}}$, option prices, and realized oil return volatilities.

Key Words: Idiosyncratic risks, jumps risks, unspanned risks, spanned risks, options on West Texas Intermediate crude oil futures

*First Draft: April, 2022. Brownbag seminar participants at Temple University provided useful feedback that I incorporated in the paper. I thank Gurdip Bakshi, Xiaohui Gao Bakshi, Yan Li, Oleg Rytchkov, Sam Rosen, and Zhaowei Zhang for many helpful conversations. All computer codes are available.

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1. Introduction

The oil price is a central focus of risk management in the commodity derivatives market. Primary oil risk concerns include price spikes and increasing volatility, both of which can be perceived as signaling economic downturns. The hedging nature of options makes them a suitable channel for understanding oil price and volatility risks, and importantly, necessitates empirically consistent oil option pricing models.

This paper shows the aversion to large upside oil price movements. The aversion is not mirrored on the downside, as evidenced by option risk premiums. The occurrence of large oil price fluctuations, which are unfavorable on the upside, motivates the theoretical option pricing with jumps. Further, the idiosyncratic volatility is modeled separately from spanned and unspanned volatility risks. Empirical consistency with my model specification, along with its performance in aligning with key variables of interest such as option prices and risk premiums, is also demonstrated.

The first phase of the paper presents two salient empirical evidence regarding crude oil price and volatility risks. I show the aversion to large upside oil price risks, as evidenced by significantly negative average excess returns from buying deep out-of-the-money (OTM) calls on oil futures in comparison to indistinguishable average excess returns from OTM puts. I corroborate the robustness of these key properties using energy options on natural gas and heating oil futures, as well as weekly crude oil option contracts.

Additionally, I show that the $\text{VIX}_t^{\text{oil}}$ is higher than the $\text{VIX}_t^{\text{S\&P500}}$ for 99% of the oil option expiration cycles, with an average $\text{VIX}_t^{\text{oil}}$ of 36.9 compared to 19.2 for $\text{VIX}_t^{\text{S\&P500}}$. As $\text{VIX}_t^{\text{S\&P500}}$ acts as a forward looking indicator for the amount of systematic volatility, this observation suggests that idiosyncratic risk is a potential influencer of oil prices. Consequently, due to their nonlinear payoffs as derivatives, the prices and risk premiums of oil options are likely to respond to the idiosyncratic risk components in the underlying asset's dynamics. My findings provide groundwork for modeling and interpreting risk ingredients.

I next consider a comprehensive model that incorporates idiosyncratic, jump, unspanned, and spanned risks to derive expressions for oil futures prices, risk-neutral futures return volatility, $\text{VIX}_t^{\text{oil}}$, and option risk premiums. Each risk is incorporated due to consideration of the empirical properties of the oil mar-

ket. The model specifies a pricing kernel where jumps can elevate marginal utilities, but the idiosyncratic component does not carry a price of risk. Unspanned risks are only covered in oil derivatives, which completes the market. The dynamics of futures, options on the oil futures, and risk premiums evolve from my specifications of the pricing kernel and the oil spot price. I show that realized return volatility, risk-neutral return volatility and $\text{VIX}_t^{\text{oil}}$ reflect an idiosyncratic component in analytic forms. Moreover, the idiosyncratic volatility along with jump and unspanned risks are relevant to negative risk premiums of OTM calls on crude oil futures, which are observed in data.

I use Kalman filtering and maximum likelihood estimation to answer three questions about the oil market: (1) Are idiosyncratic variances empirically relevant? (2) Are jump risks and/or unspanned risks priced? (3) Is there supportive evidence for model performance? This is the first time models founded in idiosyncratic risks and jump risks are being used to fit the dynamics of futures prices and oil options data.

By extracting the time-series of latent idiosyncratic and non-idiosyncratic variances, I ask additional questions: Does the model respect features of OTM option excess returns? How effective are setups that modify models with deterministic volatility to include stochastic idiosyncratic volatility and jumps? Which attributes are important for understanding the panel of option prices?

Positioning my study. To the best of my knowledge, there is no literature on whether oil models can reproduce negative risk premiums of OTM calls on crude oil futures. Existing models with deterministic futures return volatility do not account for the fact that a portion of oil return volatility and $\text{VIX}_t^{\text{oil}}$ may be linked to idiosyncratic variance. My oil model with idiosyncratic variance drivers fill these gaps.

The models of Gibson and Schwartz (1990), Schwartz (1997), and Casassus and Collin-Dufresne (2005) can be used as benchmarks for my study. Since these models abstract from stochastic volatility, they are unable to capture $\text{VIX}_t^{\text{oil}}$ or time-variation in option-mimicking portfolios for return moments. In the model of Kogan, Livdan, and Yaron (2009) and the Gaussian model of Schwartz and Smith (2000), oil prices admit deterministic return volatility. Whereas Chiang, Hughen, and Sagi (2015) and Christoffersen, Jacobs, and Li (2016) allow random volatility, none of these models incorporate idiosyncratic volatility.

My model is distinct in its inclusion of idiosyncratic and jump risks. This is different from Trolle and Schwartz (2009), who focus on model dynamics under the risk-neutral measure without jumps. I also model market incompleteness through jumps and unspanned risks in the pricing kernel dynamics, and allow idiosyncratic risk components in the dynamics of options. These distinctions are essential for capturing state-contingent risk premiums to the downside and the upside of oil futures, return volatilities, and oil option prices. Moreover, the lens of idiosyncratic risks and upside jump risks is central to understanding empirical phenomena in the oil options market, such as the time-variation in excess returns of OTM options.

The essence of my inquiry is to formalize the economic consequences of models that combine idiosyncratic, jump, unspanned, and spanned risks. I distill model structures that test for the absence of jump risks. Table 1 summarizes the modeling distinctions from the literature (I consider 14 models).

Empirical findings and model performance. The role of idiosyncratic (and jump) risks in the oil market has not been adequately explored, either in theoretical models or in empirical investigations. I evaluate the implications of the estimated model for $\mathrm{VIX}_t^{\mathrm{oil}}$, risk-neutral return volatility, and option risk premiums. Additionally, I analyze option prices based on moneyness and assess the model's performance.

I link my work to existing models of traded oil, showing how my model can replicate several data facts, such as reliably negative average excess returns of OTM calls on crude oil and insignificant average excess returns of OTM puts on crude oil. My framework allows for the pricing of unspanned volatility and jump risks, distinct from idiosyncratic risks, suggesting that idiosyncratic risks are important for describing the behaviors of crude oil contingent claims.

Benchmarking to outstanding issues, I probe empirical performance. I quantify the importance of time-varying idiosyncratic risks in two steps. First, I establish facts that connect to idiosyncratic risks. One of these data facts concerns the properties of risk-neutral return volatility. Second, to account for these stylized facts, I introduce a model that embeds idiosyncratic components. My investigation associates model suitability to alignment with conditional risk-neutral return moments.

Modeling and empirical takeaways. My model provides a framework for estimating oil models from

futures, return volatilities, and option portfolios. My analysis demonstrates that the model captures the data features of futures prices, $\text{VIX}_t^{\text{oil}}$, realized return volatilities, and options. This outcome can be attributed to the model's flexibility in incorporating idiosyncratic risks and jumps, making my framework suitable for modeling and estimating oil models using options data.

2. What data features are missing from crude oil models?

This section presents empirical properties which can help inform the development of crude oil models. I construct the time-series of excess returns of OTM puts and calls on crude oil futures over option expiration cycles and establish negative risk premiums for the upside of crude oil futures. I then analyze the behavior of risk-neutral return variance and $\text{VIX}_t^{\text{oil}}$, both of which relate to the attributes of idiosyncratic variance and are formulated from a portfolio of OTM options on crude oil futures. Finally, I explore the influence of idiosyncratic risks, an element often omitted in existing oil models, by examining whether the estimated model is able to generate the right amount of idiosyncratic and non-idiosyncratic variance to match the properties of data moments and option prices.

2.1. Data on West Texas Intermediate crude oil futures and options

I examine the data of futures and options associated with the West Texas Intermediate (WTI) crude oil, which are listed on the New York Mercantile Exchange. The WTI is a light, sweet crude oil and serves as a reference price for buyers and sellers of crude oil. The price data is provided by Barchart and includes spot, futures, and options on futures.

Let $F_t^{T_F}$ be the price of the futures contract written on the WTI crude oil that matures at T_F, and S_t the spot oil price. I employ futures with maturities of 1-, 3-, 6-, 9-, and 12 months to construct the futures curve over option expiration cycles.

I merge data on futures with option prices of different strikes, excluding in-the-money options. The near-maturity options, with an average maturity of 30 days over expiration cycles, are kept in the sample period from January 1990 to December 2019, which consists of 360 option expiration cycles. On average,

each expiration cycle contains 52 puts and 60 calls.

2.2. *Empirical support for negative (insignificant) OTM call (put) option risk premiums*

Let K be the strike price of the option written on the oil futures. OTM puts are associated with moneyness $k = \log(\frac{K}{F_t^{T_F}}) < 0$ and OTM calls are associated with $k = \log(\frac{K}{F_t^{T_F}}) > 0$. The option expires at time $t+\Delta$, with $t+\Delta < T_F$. Table 2 presents the average excess returns of options on oil futures over monthly expiration cycles, which are calculated as follows:

$$\mathbb{r}^{\text{put}}_{\{t\to t+\Delta\}}[k] = \frac{\max(K - F^{T_F}_{t+\Delta}, 0)}{\text{put}_t[K]} - R^{\text{rf}}_t, \text{ where } K \text{ corresponds to } k = -5\% \text{ and} - 10\%, \quad (1)$$

$$\underbrace{\mathbb{r}^{\text{call}}_{\{t\to t+\Delta\}}[k]}_{\text{monthly (\%)}} = \frac{\max(F^{T_F}_{t+\Delta} - K, 0)}{\text{call}_t[K]} - R^{\text{rf}}_t, \text{ where } K \text{ corresponds to } k = 5\% \text{ and } 10\%. \quad (2)$$

In equations (1)–(2), $\text{put}_t[K]$ and $\text{call}_t[K]$ are the prices of crude oil puts and calls with moneyness closest to 5% and 10% OTM at time t, respectively, and $\Delta = \frac{30}{365}$. I focus on these strike price thresholds to account for the average oil futures return volatility of 9.2% per month (32.2% annualized). Additionally,, $F^{T_F}_{t+\Delta}$ is the time $t+\Delta$ price of the futures contract, and R^{rf}_t is the gross interest rate.

Table 2 (Panel A) draws attention to the difference in average excess returns of options on oil futures of the same moneyness. Notably, the 10% OTM calls (on average, 13.9 delta) have an average excess return of -36% per *month*, compared to -22% per month for 10% OTM puts (on average, -11.6 delta).

How reliable is the observation that the average excess returns of OTM calls on crude oil futures are negative? The 95% bootstrap confidence intervals, constructed using a stationary bootstrap procedure, indicate that the average excess returns of 10% OTM calls are negative and statistically significant. The lower and upper bootstrap confidence intervals do not bracket zero. Additionally, the evidence suggests that the average excess returns of OTM puts are statistically indistinguishable from zero.

Because option risk premiums reflect the expectation of option excess returns, my findings are consistent with negative unconditional risk premiums for the upside of crude oil futures. In contrast, the unconditional risk premiums for the downside of crude oil futures are insignificant. This documented

asymmetry in state-contingent crude oil risk premiums is an important takeaway from Table 2, and can help to differentiate oil models.

Do concerns about rising oil prices become more significant at higher moneyness $k > 0$? To answer this question, I investigate the following hypotheses by jointly bootstrapping all option returns:

$$\underbrace{\overline{\mathbb{r}}^{\text{call}}_{\{t\to t+\Delta\}}[5\%] \;-\; \overline{\mathbb{r}}^{\text{call}}_{\{t\to t+\Delta\}}[10\%]}_{\text{difference in average excess returns of oil calls}} \;\geq\; 0 \qquad \text{and} \qquad \underbrace{\overline{\mathbb{r}}^{\text{put}}_{\{t\to t+\Delta\}}[5\%] \;-\; \overline{\mathbb{r}}^{\text{put}}_{\{t\to t+\Delta\}}[10\%]}_{\text{difference in average excess returns of oil puts}} \;\geq\; 0. \tag{3}$$

To assess this, I compare the empirical frequency that calls associated with an OTM moneyness k exhibit higher average returns than the one with a lower k. The resulting p-value of 0.00 (0.00) for calls (puts) indicates that risk premiums are incrementally more negative further OTM.

Conditional excess returns of options offer additional evidence on asymmetries in risk premiums of state-contingent oil claims. Following Bakshi, Crosby, and Gao (2022), I compute the average excess option returns conditional on $\{\mathcal{F}_t \in \mathfrak{s}\}$, for some information variable $\mathfrak{s}$. I focus on the following variables to represent $\mathfrak{s}$, all known at the beginning of option expiration cycles:

- *Equity VIX*: Chicago Board of Options Exchange equity volatility index ($\text{VIX}^{\text{S\&P500}}_t$).
- *Oil VIX*: Constructed from options on the WTI futures. Specifically, $\{\text{VIX}^{\text{oil}}_t\}^2 = \frac{2R^{\text{rf}}_t}{\frac{30}{365}} \int_{K<F^{T_F}_t} \frac{\text{put}_t[K]}{K^2}\, dK + \frac{2R^{\text{rf}}_t}{\frac{30}{365}} \int_{K>F^{T_F}_t} \frac{\text{call}_t[K]}{K^2}\, dK$. A high reading of $\text{VIX}^{\text{oil}}_t$ coincides with elevated uncertainty in the oil market.
- *Slope of the oil futures curve*: Positive (negative) $\log(\frac{F^{T_{F_2}}_t}{F^{T_{F_1}}_t})$ align with contango (backwardation).

To obtain option returns distinguished by economic states, I run regressions in which the excess returns of calls is the dependent variable (likewise for puts and straddle), as follows:

$$\mathbb{r}^{\text{call}}_{\{t\to t+\Delta\}}[k] \;=\; \begin{cases} \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{low}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{low}}\}} + \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{medium}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{medium}}\}} + \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{high}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{high}}\}} + \underbrace{\varepsilon_{\{t\to t+\Delta\}}}_{\text{error term}}, & \textbf{or} \\ \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{contango}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{contango}}\}} + \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{backwardation}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{backwardation}}\}} + \varepsilon_{\{t\to t+\Delta\}}. \end{cases} \tag{4}$$

The partitioned average excess returns reported in Table 2 (Panel B) differ based on $\{\mathcal{F}_t \in \mathfrak{s}\}$, yet my analysis still reveals negative conditional risk premiums of call options written on crude oil futures. Notably,

the 10% OTM calls have negative partitioned average excess returns in each state. In addition, negative risk premiums conditioning on high oil volatility states are present across OTM puts, the straddle portfolio, and OTM calls, reflecting the cost to hedge against oil volatility risks.

2.3. Excess returns of options on futures on natural gas and heating oil strengthen my evidence

To corroborate the evidence of negative average excess returns of oil call options, I examine matching evidence from two other energy markets, namely natural gas (Henry Hub) and heating oil (NY Harbor ULSD). I construct the 30 day holding period returns of 5% and 10% OTM puts and calls for each underlier over the 16-year period from January 2005 to December 2020 (191 option expiration cycles).

My analysis of Table 3 reveals three observations. Firstly, the average excess returns of OTM calls on both underliers are significantly negative. Secondly, the average excess returns of OTM put options are statistically insignificant. Finally, the average call excess returns decrease with moneyness. This suggests that market participants are consistently paying for upside protection in energy markets, which is a consistent pattern across natural gas, heating oil, and WTI. I will use this finding from the options market to investigate if it can be reconciled by crude oil models.

2.4. Risk-neutral return skewness imputed from options is positive on average and switches sign

To build evidence for idiosyncratic risks in the crude oil market, let $\mathbb{E}_t^{\mathbb{Q}}(\bullet)$ denote the risk-neutral expectation operator. I extract the skewness and kurtosis of the risk-neutral futures return distribution, for $R^{\text{oil}}_{\{t\to t+\Delta\}} \equiv \frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}$, as follows:

$$\text{Skewness}_t^{\mathbb{Q}} = \frac{\mathbb{E}_t^{\mathbb{Q}}(\{R^{\text{oil}}_{\{t\to t+\Delta\}}-1\}^3)}{\{\mathbb{E}_t^{\mathbb{Q}}(\{R^{\text{oil}}_{\{t\to t+\Delta\}}-1\}^2)\}^{3/2}} \quad \text{and} \quad \text{Kurtosis}_t^{\mathbb{Q}} = \frac{\mathbb{E}_t^{\mathbb{Q}}(\{R^{\text{oil}}_{\{t\to t+\Delta\}}-1\}^4)}{\{\mathbb{E}_t^{\mathbb{Q}}(\{R^{\text{oil}}_{\{t\to t+\Delta\}}-1\}^2)\}^{2}}, \tag{5}$$

where I apply $\mathbb{E}_t^{\mathbb{Q}}(\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}) = \mathbb{E}_t^{\mathbb{Q}}(R^{\text{oil}}_{\{t\to t+\Delta\}}) = 1$. It holds, for $n = 2,3,4$, that

$$\mathbb{E}_t^{\mathbb{Q}}(\{R^{\text{oil}}_{\{t\to t+\Delta\}}-1\}^n) = \int_{K<F_t^{T_F}} \mathbb{w}[K]\,\text{put}_t[K]\,dK + \int_{K>F_t^{T_F}} \mathbb{w}[K]\,\text{call}_t[K]\,dK, \text{ with } \mathbb{w}[K] = \frac{(n-1)n(\frac{K}{F_t^{T_F}}-1)^{n-2}}{(F_t^{T_F})^2}. \tag{6}$$

Table 4 reports the summary statistics on extracted risk-neutral return skewness and kurtosis, together with the risk-neutral return volatility (i.e., $\sqrt{\mathbb{E}_t^{\mathbb{Q}}(\{R^{\text{oil}}_{\{t\to t+\Delta\}}-1\}^2)}$) and $\text{VIX}_t^{\text{oil}}$.

Two observations from option prices are relevant to oil models. Firstly, the time-varying effects in risk-neutral return moments and $\text{VIX}_t^{\text{oil}}$ cannot be explained by models of oil futures with deterministic return volatility. Secondly, the presence of idiosyncratic risks is indicated by the average $\text{VIX}_t^{\text{oil}}$ ($\text{VIX}_t^{\text{S\&P500}}$) being 36.9% (19.2%). The estimate of $\mu^\star$ in the regression $\log(\frac{\text{VIX}_t^{\text{oil}}}{\text{VIX}_t^{\text{S\&P500}}}) = \mu^\star + \widetilde{e}_t$ is 0.66 (NW[p] of 0.00), which implies that $\text{VIX}_t^{\text{oil}}$ is significantly higher than $\text{VIX}_t^{\text{S\&P500}}$. The two time-series comove imperfectly: The $\text{VIX}_t^{\text{oil}}$ (changes in $\text{VIX}_t^{\text{oil}}$) and $\text{VIX}_t^{\text{S\&P500}}$ (changes in $\text{VIX}_t^{\text{S\&P500}}$) exhibit a correlation of 0.62 (0.34).

The mean $\text{Skewness}_t^{\mathbb{Q}}$ is significantly positive (based on the bootstrap evidence), and $\text{Skewness}_t^{\mathbb{Q}}$ can change from positive to negative with a frequency of 14%. Furthermore, the high mean $\text{Kurtosis}_t^{\mathbb{Q}}$ reinforces the presence of idiosyncratic risks. In Appendix B, I consider model restrictions under which the properties of idiosyncratic risks can lead to sign switching risk-neutral return skewness, as well as risk-neutral kurtosis in excess of three.

2.5. Summary of my findings from options on oil futures and missing modeling elements

My empirical findings encapsulate distinct aspects. Firstly, Gibson and Schwartz (1990) and Schwartz (1997) consider models driven by the logarithm of the spot price and the cost of carry, which are absent of stochastic futures return volatility and imply a deterministic $\text{VIX}_t^{\text{oil}}$. Models of crude oil futures with deterministic volatility — including Casassus and Collin-Dufresne (2005), Cortazar and Lorenzo (2006), Kogan, Livdan, and Yaron (2009), and Heath (2019) — are inconsistent with dynamic movements in $\text{VIX}_t^{\text{oil}}$ and not suitable for generating sign-reversals in the risk-neutral return skewness.

The approach in Trolle and Schwartz (2009) accommodates unspanned stochastic volatility under the risk-neutral measure but abstains from modeling idiosyncratic components in the oil price dynamics or price jumps. Moreover, their work does not address option risk premiums. I equip oil price dynamics with idiosyncratic risks and jump risks and conceptualize their implications for understanding futures return volatility and $\text{VIX}_t^{\text{oil}}$, as well as expected excess returns of options on crude oil futures.

Chiang, Hughen, and Sagi (2015) model volatility, but do not develop a role for idiosyncratic risks nor examine returns of oil options or formalize option risk premiums. This is distinct from Christoffersen, Jacobs, and Pan (2022) and Jacobs and Li (2023), as my focus is on a parametric model with two added elements: (i) my model can reconcile the negative risk premiums of OTM calls (with small delta), and (ii) I emphasize empirical model performance. The framework of Crosby and Frau (2022) does not investigate option risk premiums nor incorporate idiosyncratic risks, while Cheng, Tang, and Yan (2022) study the impact of hedgers and speculators on the slope of the commodity implied volatility curves.

Equilibrium models of the oil market (e.g., Routledge, Seppi, and Spatt (2000), Carlson, Khokher, and Titman (2007), Baker and Routledge (2017), Ready (2018), and Bornstein, Krusell, and Rebelo (2023)) do not connect to the pattern of option returns and option risk premiums, nor do they assign a role to idiosyncratic volatility. However, my analysis finds that idiosyncratic volatility can help to explain the behavior of oil futures return volatilities (both realized and risk-neutral).

To conclude, extant specifications forgo the modeling of both idiosyncratic and jump risks, do not deduce the implications of $\text{VIX}_t^{\text{oil}}$ and risk-neutral return moments, and available models do not reconcile option risk premiums. These features motivate my modeling and empirical assessment exercises.

Many studies, notably Kang, Rouwenhorst, and Tang (2014) (David (2019)) investigate commodity (energy) risk premiums but do not characterize risk premiums to the return downside versus the return upside. While both the 10% down and up futures return movements are low probability events over monthly oil option expiration cycles, my novelty lies in disentangling option risk premiums. Oil models should be aligned with the expectation of option payoffs under both the real-world and the risk-neutral probability measures. Hence, oil option risk premiums are critical quantities to model and understand empirically.

3. Oil model with idiosyncratic, jump, unspanned, and spanned risks

I focus on a framework that incorporates salient characteristics of the oil market. By considering the relevance of idiosyncratic risks and jump risks, I invite models that could fall in the intersection of high

$\mathrm{VIX}_t^{\mathrm{oil}}$ and plausible option price dynamics.

My inquiry formalizes a model with idiosyncratic and jump risks to ask: How far can the framework go in explaining the data on $\mathrm{VIX}_t^{\mathrm{oil}}$, realized oil futures return volatilities, and options on oil futures?

Let $\mathbb{P}$ denote the real-world probability measure. Since markets are not assumed complete, there is neither a unique risk-neutral measure nor a unique pricing kernel. I consider a risk-neutral measure $\mathbb{Q}$ and a pricing kernel M_t consistent with no arbitrage. Let $\mathbb{E}_t^{\mathbb{P}}(\bullet) \equiv \mathbb{E}^{\mathbb{P}}(\bullet|\mathcal{F}_t)$ ($\mathbb{E}_t^{\mathbb{Q}}(\bullet) \equiv \mathbb{E}^{\mathbb{Q}}(\bullet|\mathcal{F}_t)$) be the expectation under $\mathbb{P}$ ($\mathbb{Q}$), conditional on $\mathcal{F}_t$. Furthermore, r_t is the spot interest rate, assumed deterministic, and S_t is the spot price of crude oil. I consider the dynamics of M_t and S_t as follows:

$$\frac{dM_t}{M_{t-}} = -r_t\,dt + \mho[\mathrm{V}_t]\underbrace{d\mathbb{z}_t^{\mathbb{P}}}_{\text{spanned risks}} + \Pi[\mathrm{V}_t]\underbrace{d\mathbb{u}_t^{\mathbb{P}}}_{\text{unspanned risks}} + \{\underbrace{(e^{J_t}-1)d\mathbb{N}_t^{\mathbb{P}}}_{\text{jump risks}} - \underbrace{\lambda_{\mathrm{J}}^{\mathbb{P}}\,\mathbb{E}_t^{\mathbb{P}}(e^{J_t}-1)dt}_{\text{compensator}}\}, \tag{7}$$

$$\mho[\mathrm{V}_t] = -\chi_{\mathrm{v}}^*\sqrt{\mathrm{V}_t} \quad \text{and} \quad \Pi[\mathrm{V}_t] = -\chi_{\mathrm{u}}^*\sqrt{\mathrm{V}_t}, \tag{8}$$

$$\underbrace{d\mathbb{N}_t^{\mathbb{P}}}_{\text{Poisson jump}} = \begin{cases} 1 & \text{with probability } \lambda_{\mathrm{J}}^{\mathbb{P}}\,dt \\ 0 & \text{with probability } 1-\lambda_{\mathrm{J}}^{\mathbb{P}}\,dt, \end{cases} \quad \text{where } J_t \sim \underbrace{\mathrm{Normal}(\mu_{\mathrm{J}}^{\mathbb{P}}, \{\sigma_{\mathrm{J}}^{\mathbb{P}}\}^2)}_{\text{oil jump size distribution}}, \quad \text{and} \tag{9}$$

$$\frac{dS_t}{S_{t-}} = (\underbrace{\mathbf{y}_t}_{\text{cost of carry}} - \frac{1}{dt}\mathrm{cov}_t^{\mathbb{P}}(\frac{dM_t}{M_{t-}}, \frac{dS_t}{S_{t-}}))\,dt + \sqrt{\mathrm{V}_t}\underbrace{d\mathbb{z}_t^{\mathbb{P}}}_{\text{spanned risks}} + \sqrt{\mathrm{I}_t}\underbrace{d\varepsilon_t^{\mathbb{P}}}_{\text{idiosyncratic risks}} + \{\underbrace{(e^{J_t}-1)d\mathbb{N}_t^{\mathbb{P}}}_{\text{jump risks}} - \lambda_{\mathrm{J}}^{\mathbb{P}}\,\mathbb{E}_t^{\mathbb{P}}(e^{J_t}-1)dt\}, \tag{10}$$

where I_t (V_t) categorizes the idiosyncratic (non-idiosyncratic) diffusive variance component of $\frac{dS_t}{S_{t-}}$.

There are four distinct sources of risks. Specifically, idiosyncratic risks, which are detached from the dynamics of the pricing kernel, are represented by the standard Brownian motion $\varepsilon_t^{\mathbb{P}}$. Additionally, oil price discontinuities are captured through a Poisson random variable $d\mathbb{N}_t^{\mathbb{P}}$ with intensity $\lambda_{\mathrm{J}}^{\mathbb{P}}$, where the jump size J_t is assumed to be IID normal, following Merton (1976).

My framework captures unspanned risks, which are represented by the standard Brownian motion $\mathbb{u}_t^{\mathbb{P}}$, and are distinguished from $\varepsilon_t^{\mathbb{P}}$ by their effect on the dynamics of the pricing kernel. In economic terms, $\mathbb{u}_t^{\mathbb{P}}$ risks also impact the evolution of the non-idiosyncratic variance component V_t (as noted in (12)). Spanned risks are modeled by the standard Brownian motion $\mathbb{z}_t^{\mathbb{P}}$, and introduce a covariation between $\frac{dS_t}{S_{t-}}$ and $d\mathrm{V}_t$.

In light of (7)–(8) and constants χ_{v}^{*}, and χ_{u}^{*} , the measure $\mathbb{Q}$ is linked to the measure $\mathbb{P}$, as follows:

$$d\mathbb{z}_t^{\mathbb{P}} \;=\; d\mathbb{z}_t^{\mathbb{Q}} \;+\; \mho[\mathrm{V}_t]\,dt, \qquad d\mathbb{u}_t^{\mathbb{P}} \;=\; d\mathbb{u}_t^{\mathbb{Q}} \;+\; \Pi[\mathrm{V}_t]\,dt \qquad \text{and} \quad d\varepsilon_t^{\mathbb{P}} \;=\; d\varepsilon_t^{\mathbb{Q}}. \tag{11}$$

Equation (11) parameterizes the compensation for spanned and unspanned risks, while idiosyncratic risks are unpriced. All sources of risks are uncorrelated. For tractability of pricing relationship, I assume that

$$d\mathrm{V}_t \;=\; (\theta_{\mathrm{v}}^{\mathbb{P}} - \kappa_{\mathrm{v}}^{\mathbb{P}}\,\mathrm{V}_t)\,dt \;+\; \sigma_{\mathrm{v}}\,\rho_{\mathrm{v}}\,\sqrt{\mathrm{V}_t}\;\underbrace{d\mathbb{z}_t^{\mathbb{P}}}_{\text{spanned risks}} \;+\; \sigma_{\mathrm{v}}\sqrt{1-\rho_{\mathrm{v}}^2}\,\sqrt{\mathrm{V}_t}\;\underbrace{d\mathbb{u}_t^{\mathbb{P}}}_{\text{unspanned risks}}, \qquad \text{and} \tag{12}$$

$$d\mathrm{I}_t \;=\; (\theta_{\mathrm{I}}^{\mathbb{P}} - \kappa_{\mathrm{I}}^{\mathbb{P}}\,\mathrm{I}_t)\,dt \;+\; \sigma_{\mathrm{I}}\,\sqrt{\mathrm{I}_t}\,d\varepsilon_t^{\mathbb{P}}. \tag{13}$$

Employing (11) in conjunction with (12) and (13), $\frac{dS_t}{S_{t-}}$ obeys the following dynamics under $\mathbb{Q}$:

$$\frac{dS_t}{S_{t-}} \;=\; \mathbf{y}_t\,dt \;+\; \sqrt{\mathrm{V}_t}\,d\mathbb{z}_t^{\mathbb{Q}} \;+\; \sqrt{\mathrm{I}_t}\,d\varepsilon_t^{\mathbb{Q}} \;+\; \{(e^{J_t}-1)d\mathbb{N}_t^{\mathbb{Q}} \;-\; \lambda_{\mathrm{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1)dt\}, \quad \text{where,} \tag{14}$$

$$d\mathrm{V}_t \;=\; (\theta_{\mathrm{v}}^{\mathbb{Q}} - \kappa_{\mathrm{v}}^{\mathbb{Q}}\,\mathrm{V}_t)\,dt + \sigma_{\mathrm{v}}\,\rho_{\mathrm{v}}\,\sqrt{\mathrm{V}_t}\,d\mathbb{z}_t^{\mathbb{Q}} + \sigma_{\mathrm{v}}\sqrt{1-\rho_{\mathrm{v}}^2}\,\sqrt{\mathrm{V}_t}\,d\mathbb{u}_t^{\mathbb{Q}}, \qquad \text{and} \tag{15}$$

$$\kappa_{\mathrm{v}}^{\mathbb{Q}} \;\equiv\; \kappa_{\mathrm{v}}^{\mathbb{P}} \;+\; \underbrace{\chi_{\mathrm{v}}^{*}\,\sigma_{\mathrm{v}}\,\rho_{\mathrm{v}}}_{\text{risk premium}} \;+\; \underbrace{\chi_{\mathrm{u}}^{*}\,\sigma_{\mathrm{v}}\sqrt{1-\rho_{\mathrm{v}}^2}}_{\text{risk premium}}, \qquad \text{and} \qquad \theta_{\mathrm{v}}^{\mathbb{Q}} \;\equiv\; \theta_{\mathrm{v}}^{\mathbb{P}}. \tag{16}$$

Recognizing that $\mathbb{E}_t^{\mathbb{P}}(\frac{dS_t}{S_{t-}}) - \mathbf{y}_t dt = -\mathrm{cov}_t^{\mathbb{P}}(\frac{dM_t}{M_{t-}}, \frac{dS_t}{S_{t-}})$, for *spot cost of carry*, denoted by $\mathbf{y}_t$, I have $F_t^{T_F} = \mathbb{E}_t^{\mathbb{Q}}(S_{T_F}) = S_t \times \mathbb{E}_t^{\mathbb{Q}}(e^{\int_t^{T_F}\mathbf{y}_u\,du})$. The last equality follows from the drift restriction in equation (14).

One may view $\mathbf{y}_t$ as reflecting the combined effect of interest rates, storage, and convenience yield (e.g., Fama and French (1988)). I consider the following candidate specification for $\mathbf{y}_t$ in modeling:

$$\underbrace{\mathbf{y}_t}_{\text{cost of carry}} \;=\; \delta_{\mathbf{y}}^{\circ} \;+\; \delta_{\mathrm{I}}^{*}\times\mathrm{I}_t \;+\; \delta_{\mathrm{v}}^{*}\times\mathrm{V}_t, \quad \text{(spot cost of carry assumption for valuing oil futures)} \tag{17}$$

where $\delta_{\mathbf{y}}^{\circ}$, δ_{I}^{*}, and δ_{v}^{*} are constants. The pricing of oil-linked contingent claims is then based on exploiting the dynamics of S_t, V_t, and I_t. To evaluate the applicability of these pricing relations, I empirically study a panel of options on crude oil futures.

Result 1 (Model of oil futures prices) Solving $F_t^{T_F} = S_t \times \mathbb{E}_t^{\mathbb{Q}}(e^{\int_t^{T_F}\mathbf{y}_u\,du})$, it holds that

$$\frac{F_t^{T_F}}{S_t} \;=\; \exp\big(\mathbb{a}_{\mathrm{fut}}[\tau] \;+\; \mathbb{b}_{\mathrm{fut}}[\tau]\times\mathrm{V}_t \;+\; \mathbb{c}_{\mathrm{fut}}[\tau]\times\mathrm{I}_t\big), \qquad \text{with } \tau \;\equiv\; T_F - t. \tag{18}$$

The coefficients $\mathbb{a}_{\text{fut}}[\tau]$, $\mathbb{b}_{\text{fut}}[\tau]$, and $\mathbb{c}_{\text{fut}}[\tau]$ are shown in (A1)–(A9) of Appendix A. ■

The approach of specifying the spot cost of carry $\mathbf{y}_t$ — as opposed to an alternative in Heath, Jarrow, and Morton (1992) — is favored by several studies, including the model class in Schwartz (1997), Yan (2002), Casassus and Collin-Dufresne (2005), Casassus, Liu, and Tang (2013), and Chiang, Hughen, and Sagi (2015). Intrinsic to $F_t^{T_F}$ dynamics in my setup is the influence of idiosyncratic, unspanned, spanned, as well as jump risks. Thus, by applying Ito's lemma, I derive the following dynamics:

$$\begin{aligned}
\frac{dF_t^{T_F}}{F_{t-}^{T_F}} &= (-\frac{1}{dt}\text{cov}_t^{\mathbb{P}}(\frac{dM_t}{M_t}, \frac{dF_t^{T_F}}{F_t^{T_F}}))\,dt \;+\; \{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}\sqrt{\text{V}_t}\,d\mathbb{z}_t^{\mathbb{P}} \;+\; \mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\sqrt{\text{V}_t}\,d\mathbb{u}_t^{\mathbb{P}} \\
&\quad + \{1+\mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\text{I}}\}\sqrt{\text{I}_t}\,d\varepsilon_t^{\mathbb{P}} \;+\; \{(e^{J_t}-1)d\mathbb{N}_t^{\mathbb{P}} \;-\; \lambda_{\text{J}}^{\mathbb{P}}\,\mathbb{E}_t^{\mathbb{P}}(e^{J_t}-1)dt\}, \qquad \text{under } \mathbb{P} \quad \text{and} \qquad (19) \\
\frac{dF_t^{T_F}}{F_{t-}^{T_F}} &= \{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\,\rho_{\text{v}}\}\sqrt{\text{V}_t}\,d\mathbb{z}_t^{\mathbb{Q}} \;+\; \mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\sqrt{\text{V}_t}\,d\mathbb{u}_t^{\mathbb{Q}} \\
&\quad + \{1+\mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\text{I}}\}\sqrt{\text{I}_t}\,d\varepsilon_t^{\mathbb{Q}} \;+\; \{(e^{J_t}-1)d\mathbb{N}_t^{\mathbb{Q}} \;-\; \lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1)dt\} \qquad \text{under } \mathbb{Q}. \qquad (20)
\end{aligned}$$

The modeling of futures price growth in equation (19) reflects the idea that oil futures price dynamics may not be isolated from unspanned risks. This is due to $\mathbb{u}_t^{\mathbb{P}}$ risks influencing V_t dynamics.

In the manner of pricing the log contract, $\{\text{VIX}_t^{\text{oil}}\}^2 = \mathbb{E}_t^{\mathbb{Q}}(\{-\frac{2}{\Delta}\}\log(\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}))$, for $\Delta = \frac{30}{365}$.

Result 2 ($\text{VIX}_t^{\text{oil}}$ has an idiosyncratic variance component) I derive as follows:

$$\begin{aligned}
\{\text{VIX}_t^{\text{oil}}\}^2 &= \underbrace{2\lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1-J_t)}_{\text{jump contribution}} \;+\; \frac{1}{\Delta}(\{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}^2+\{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2)\,\mathbb{E}_t^{\mathbb{Q}}(\int_t^{t+\Delta}\text{V}_\ell\,d\ell) \\
&\quad + \frac{1}{\Delta}\{1 \;+\; \mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\text{I}}\}^2\,\mathbb{E}_t^{\mathbb{Q}}(\int_t^{t+\Delta}\text{I}_\ell\,d\ell) \qquad (21) \\
&= \mathbb{a}_{\text{vix}}[\Delta] \;+\; \mathbb{b}_{\text{vix}}[\Delta]\times\text{V}_t \;+\; \mathbb{c}_{\text{vix}}[\Delta]\times\text{I}_t, \qquad (22)
\end{aligned}$$

where $\mathbb{a}_{\text{vix}}[\Delta]$, $\mathbb{b}_{\text{vix}}[\Delta]$, and $\mathbb{c}_{\text{vix}}[\Delta]$ are presented in (A13)–(A15) of Appendix A. ■

My framework reveals that $\text{VIX}_t^{\text{oil}}$ contains a component due to idiosyncratic variance I_t. Additionally, the performance of alternative $\text{VIX}_t^{\text{oil}}$ models has yet to be examined for estimated models (e.g., Trolle and Schwartz (2009, 2010), Chiang, Hughen, and Sagi (2015), and Christoffersen, Jacobs, and Li (2016)). Thus, my approach and assessments can assist with model selection and differentiation.

Result 3 (A function of risk-neutral futures return variance is linear in V_t and I_t) It holds that

$$\log(1 + \underbrace{\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}} - 1\}^2)}_{\text{oil futures return variance under } \mathbb{Q}}) = \mathbb{a}_{\text{vol}}[\Delta] + \mathbb{b}_{\text{vol}}[\Delta] \times \mathrm{V}_t + \mathbb{c}_{\text{vol}}[\Delta] \times \mathrm{I}_t, \tag{23}$$

where $\mathbb{a}_{\text{vol}}[\Delta]$, $\mathbb{b}_{\text{vol}}[\Delta]$, and $\mathbb{c}_{\text{vol}}[\Delta]$ are presented in (A37) of Appendix A. ■

The characterization in (23) conveys that a function of risk-neutral return variance is linear in idiosyncratic variance, and this feature facilitates model estimation. I operationalize this measurement equation step by inferring $\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}} - 1\}^2)$ from a portfolio of OTM puts and calls on crude oil futures. Specifically, I employ equation (6) and set $n = 2$.

In order to demonstrate the role of idiosyncratic risks in the crude oil market, the instantaneous variance of log futures returns under $\mathbb{P}$ can be broken down into three parts as follows:

$$\begin{aligned} \mathbb{v}_t^{\mathbb{P}} \equiv \frac{1}{dt}\mathrm{var}_t^{\mathbb{P}}[d\log(F_t^{T_F})] &= \lambda_{\mathrm{J}}^{\mathbb{P}}(\{\mu_{\mathrm{J}}^{\mathbb{P}}\}^2 + \{\sigma_{\mathrm{J}}^{\mathbb{P}}\}^2) \\ &+ (\{1 + \mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\mathrm{V}}\,\rho_{\mathrm{V}}\}^2 + \{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\mathrm{V}}\sqrt{1-\rho_{\mathrm{V}}^2}\}^2) \times \mathrm{V}_t \\ &+ \{1 + \mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\mathrm{I}}\}^2 \times \mathrm{I}_t. \end{aligned} \tag{24}$$

I define *expected integrated variance* as $\mathrm{rv}_{\{t\to t+\Delta\}}^{\text{futures},\mathbb{P}} \equiv \mathbb{E}_t^{\mathbb{P}}(\int_t^{t+\Delta} \mathbb{v}_\ell^{\mathbb{P}}\, d\ell)$ and $\mathrm{rv}_{\{t\to t+\Delta\}}^{\text{spot},\mathbb{P}} = \mathrm{rv}_{\{t\to t+\Delta\}}^{\text{futures},\mathbb{P}}\Big|_{T_F - t = 0}$.

Result 4 (Expected integrated variance of futures returns under $\mathbb{P}$) The following relations hold:

$$\text{Spot return variance under } \mathbb{P}: \quad \mathrm{rv}_{\{t\to t+\Delta\}}^{\text{spot},\mathbb{P}} = \mathbb{a}_{\text{rv}}^{\text{spot}}[\Delta] + \mathbb{b}_{\text{rv}}^{\text{spot}}[\Delta] \times \mathrm{V}_t + \mathbb{c}_{\text{rv}}^{\text{spot}}[\Delta] \times \mathrm{I}_t, \tag{25}$$

$$\text{Futures return variance under } \mathbb{P}: \quad \mathrm{rv}_{\{t\to t+\Delta\}}^{\text{futures},\mathbb{P}} = \mathbb{a}_{\text{rv}}^{\text{fut}}[\Delta] + \mathbb{b}_{\text{rv}}^{\text{fut}}[\Delta] \times \mathrm{V}_t + \mathbb{c}_{\text{rv}}^{\text{fut}}[\Delta] \times \mathrm{I}_t, \tag{26}$$

where $\mathbb{a}_{\text{rv}}^{\text{fut}}[\Delta]$, $\mathbb{b}_{\text{rv}}^{\text{fut}}[\Delta]$, and $\mathbb{c}_{\text{rv}}^{\text{fut}}[\Delta]$ (respectively, $\mathbb{a}_{\text{rv}}^{\text{spot}}[\Delta]$, $\mathbb{b}_{\text{rv}}^{\text{spot}}[\Delta]$, and $\mathbb{c}_{\text{rv}}^{\text{spot}}[\Delta]$) are presented in (A38)–(A40) (respectively, (A41)–(A42)) of Appendix A. ■

Finally, I denote the *variance spreads* — that reflect the difference between $\mathbb{P}$ and $\mathbb{Q}$ measure quantities — as $\mathrm{VSPREAD}_{\{t\to t+\Delta\}} \equiv \mathrm{rv}_{\{t\to t+\Delta\}}^{\text{futures},\mathbb{P}} - \Delta \times \{\mathrm{VIX}_t^{\text{oil}}\}^2$.

Result 5 (Variance spreads) It holds that

$$\text{VSPREAD}_{\{t\to t+\Delta\}} \;=\; \mathbb{a}_{\text{vspread}}[\Delta] \;+\; \mathbb{b}_{\text{vspread}}[\Delta] \times \text{V}_t. \tag{27}$$

The variance spreads — an analog to the variance risk premiums — exhibit a zero exposure to idiosyncratic variance. I present $\mathbb{a}_{\text{vspread}}[\Delta]$ and $\mathbb{b}_{\text{vspread}}[\Delta]$ in (A44)–(A45) of Appendix A. ■

Having derived theoretical relations applicable to measurements in the Kalman filter, I proceed to estimate the model and extract V_t and I_t that are consistent with options, futures, and variance data. Notably, my framework recognizes that OTM call option risk premiums can be negative, and $\text{VIX}_t^{\text{oil}}$ as well as risk-neutral futures return volatility can accommodate the workings of time-varying idiosyncratic variance.

4. Reconciling empirical properties of option prices and call risk premiums

My investigation is centered on the role of idiosyncratic risks and jump risks in the context of estimated oil models. Pertinent to model estimation using maximum-likelihood and Kalman filtering, the measurement equations formalize the association between the data constructs and unobservable V_t and I_t.

I postulate the model in state-space form, as follows:

$$\underset{(5\times 1)}{\mathbf{Y}_t} \equiv \underbrace{\begin{bmatrix} \log(\frac{F_t^{T_F}}{S_t}) \\ \log(1+\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}-1\}^2)) \\ \text{rv}_{\{t\to t+\Delta\}}^{\text{spot},\mathbb{P}} \\ \text{rv}_{\{t\to t+\Delta\}}^{\text{futures},\mathbb{P}} \\ \text{VSPREAD}_{\{t\to t+\Delta\}} \end{bmatrix}}_{\text{all data in monthly units}} = \underbrace{\begin{bmatrix} \mathbb{a}_{\text{fut}}[\tau] & \mathbb{b}_{\text{fut}}[\tau] & \mathbb{c}_{\text{fut}}[\tau] \\ \mathbb{a}_{\text{vol}}[\Delta] & \mathbb{b}_{\text{vol}}[\Delta] & \mathbb{c}_{\text{vol}}[\Delta] \\ \mathbb{a}_{\text{rv}}^{\text{spot}}[\Delta] & \mathbb{b}_{\text{rv}}^{\text{spot}}[\Delta] & \mathbb{c}_{\text{rv}}^{\text{spot}}[\Delta] \\ \mathbb{a}_{\text{rv}}^{\text{fut}}[\Delta] & \mathbb{b}_{\text{rv}}^{\text{fut}}[\Delta] & \mathbb{c}_{\text{rv}}^{\text{fut}}[\Delta] \\ \mathbb{a}_{\text{vspread}}[\Delta] & \mathbb{b}_{\text{vspread}}[\Delta] & \boxed{0} \end{bmatrix}}_{\equiv\, \mathbb{C}[\Theta]} \underbrace{\begin{bmatrix} 1 \\ \text{V}_t \\ \text{I}_t \end{bmatrix}}_{\mathbf{X}_t} + \widetilde{\boldsymbol{\varepsilon}}_t, \qquad \widetilde{\boldsymbol{\varepsilon}}_t \sim \mathcal{N}(0, \boldsymbol{\Omega}). \tag{28}$$

The parametrization $\mathbf{Y}_t = \mathbb{C}[\Theta]\,\mathbf{X}_t + \widetilde{\boldsymbol{\varepsilon}}_t$ acknowledges imperfect modeling, Θ is the vector of parameters to be estimated, and $\widetilde{\boldsymbol{\varepsilon}}_t$ is a vector of i.i.d. Gaussian errors with constant (diagonal) covariance matrix Ω. Additionally, I use matching data on the spot crude oil and futures prices from Barchart.

While the transition density of $\begin{bmatrix} \text{V}_t \\ \text{I}_t \end{bmatrix}$ is unknown, its conditional mean and variance admit a closed-form. I approximate the transition density with a Gaussian density with the same conditional mean and variance. The transition equations are as follows:

$$\begin{bmatrix} \mathrm{V}_t \\ \mathrm{I}_t \end{bmatrix} = \begin{bmatrix} \frac{\theta_{\mathrm{v}}^{\mathbb{P}}}{\kappa_{\mathrm{v}}^{\mathbb{P}}}(1-e^{-\kappa_{\mathrm{v}}^{\mathbb{P}}\Delta}) \\ \frac{\theta_{\mathrm{I}}^{\mathbb{P}}}{\kappa_{\mathrm{I}}^{\mathbb{P}}}(1-e^{-\kappa_{\mathrm{I}}^{\mathbb{P}}\Delta}) \end{bmatrix} + \begin{bmatrix} e^{-\kappa_{\mathrm{v}}^{\mathbb{P}}\Delta} & 0 \\ 0 & e^{-\kappa_{\mathrm{I}}^{\mathbb{P}}\Delta} \end{bmatrix} \begin{bmatrix} \mathrm{V}_{t-1} \\ \mathrm{I}_{t-1} \end{bmatrix} + \widetilde{\boldsymbol{\eta}}_t, \qquad \widetilde{\boldsymbol{\eta}}_t \sim \mathcal{N}(0, \boldsymbol{\Lambda}_t), \tag{29}$$

where $\boldsymbol{\Lambda}_t$ is diagonal with elements $\frac{\sigma_{\mathrm{v}}^2}{\kappa_{\mathrm{v}}^{\mathbb{P}}}\{e^{-\kappa_{\mathrm{v}}^{\mathbb{P}}\Delta} - e^{-2\kappa_{\mathrm{v}}^{\mathbb{P}}\Delta}\}\mathrm{V}_{t-1} + \frac{\theta_{\mathrm{v}}^{\mathbb{P}}\sigma_{\mathrm{v}}^2}{2\{\kappa_{\mathrm{v}}^{\mathbb{P}}\}^2}(1-e^{-\kappa_{\mathrm{v}}^{\mathbb{P}}\Delta})^2$ and $\frac{\sigma_{\mathrm{I}}^2}{\kappa_{\mathrm{I}}^{\mathbb{P}}}\{e^{-\kappa_{\mathrm{I}}^{\mathbb{P}}\Delta} - e^{-2\kappa_{\mathrm{I}}^{\mathbb{P}}\Delta}\}\mathrm{I}_{t-1} + \frac{\theta_{\mathrm{I}}^{\mathbb{P}}\sigma_{\mathrm{I}}^2}{2\{\kappa_{\mathrm{I}}^{\mathbb{P}}\}^2}(1-e^{-\kappa_{\mathrm{I}}^{\mathbb{P}}\Delta})^2$. I estimate the model using data on futures prices, risk-neutral return variances extracted from options, realized variances of futures and spot returns, and variance spreads.

My design of including options along with realized variances disciplines measurements. In particular, it helps to distinguish between jump parameters under $\mathbb{P}$ and $\mathbb{Q}$ by looking at quantities that are sensitive to jumps. Additionally, I take advantage of the feature that I_t and variance spreads are uncorrelated, which isolates V_t separately from I_t.

Jump risks may necessitate risk premiums. For instance, futures prices jump according to a Poisson process with intensity $\lambda_{\mathrm{J}}^{\mathbb{Q}}$ under $\mathbb{Q}$. Valuing parsimony, I consider the ensuing links under $\mathbb{P}$ and $\mathbb{Q}$:

$$\lambda_{\mathrm{J}}^{\mathbb{Q}} = \lambda_{\mathrm{J}}^{\mathbb{P}} + \{\lambda_{\mathrm{J}}^{\mathrm{rp}}\}^2, \qquad \text{(risk premium for jump intensity)} \tag{30}$$

$$\mu_{\mathrm{J}}^{\mathbb{Q}} = \mu_{\mathrm{J}}^{\mathbb{P}} \times \exp(\mu_{\mathrm{J}}^{\mathrm{rp}}), \qquad \text{(risk premium for (absolute) mean jump size)} \tag{31}$$

$$\{\sigma_{\mathrm{J}}^{\mathbb{Q}}\}^2 = \{\sigma_{\mathrm{J}}^{\mathbb{P}}\}^2 \times \exp(\sigma_{\mathrm{J}}^{\mathrm{rp}}). \qquad \text{(risk premium for variance of jump size)} \tag{32}$$

Distilling implications for the pricing kernel of the oil market in an incomplete markets setting, I can evaluate whether $\lambda_{\mathrm{J}}^{\mathrm{rp}} = 0$, $\mu_{\mathrm{J}}^{\mathrm{rp}} = 0$, and $\sigma_{\mathrm{J}}^{\mathrm{rp}} = 0$.

My approach leads us to estimate 23 parameters:

$$\underset{(23\times 1)}{\Theta} = [\underbrace{\kappa_{\mathrm{I}}^{\mathbb{P}}, \theta_{\mathrm{I}}^{\mathbb{P}}, \sigma_{\mathrm{I}}}_{\text{idiosyncratic}}, \underbrace{\mu_{\mathrm{J}}^{\mathbb{P}}, \sigma_{\mathrm{J}}^{\mathbb{P}}, \lambda_{\mathrm{J}}^{\mathbb{P}}, \mu_{\mathrm{J}}^{\mathrm{rp}}, \sigma_{\mathrm{J}}^{\mathrm{rp}}, \lambda_{\mathrm{J}}^{\mathrm{rp}}}_{\text{jump}}, \underbrace{\delta_{\mathbf{y}}^{\circ}, \delta_{\mathrm{I}}^{*}, \delta_{\mathrm{v}}^{*}}_{\text{cost of carry}}, \underbrace{\kappa_{\mathrm{v}}^{\mathbb{P}}, \theta_{\mathrm{v}}^{\mathbb{P}}, \sigma_{\mathrm{v}}, \chi_{\mathrm{v}}^{*}, \chi_{\mathrm{u}}^{*}, \rho_{\mathrm{v}}}_{\text{non-idiosyncratic}}, \underbrace{\Omega_1, \Omega_2, \Omega_3, \Omega_4, \Omega_5}_{\text{measurement}}]. \tag{33}$$

The Kalman filter produces one-step-ahead forecasts for $\mathbf{Y}_t$, denoted by $\widehat{\mathbf{Y}}_{t|t-1}$, and the corresponding error covariance matrices, $\boldsymbol{\Sigma}_{t|t-1}$. In this case, the log-likelihood function can be constructed as follows:

$$\mathbb{L}[\Theta] = -\frac{1}{2}\sum_{t=1}^{T}\Big\{5\times\log(2\pi) + \log(\boldsymbol{\Sigma}_{t|t-1}) + (\mathbf{Y}_t - \widehat{\mathbf{Y}}_{t|t-1})'\boldsymbol{\Sigma}_{t|t-1}^{-1}(\mathbf{Y}_t - \widehat{\mathbf{Y}}_{t|t-1})\Big\}, \tag{34}$$

where T is the number of expiration cycles. The quasi-maximum-likelihood estimator, Θ, is $\widehat{\Theta} = \arg\max_{\Theta} \mathbb{L}[\Theta]$.

An essential question is: Does the estimated model generate the correct properties of V_t and I_t? To answer this, Table 5 examines ten quantities and provides a comparison of model fit versus actual. Since assessing a model based on its average fit could hide its incapability to capture dynamic aspects of the oil market, I present consistency across the minimum, percentiles, and the maximum.

4.1. My analysis supports the economic relevance of idiosyncratic variance

My objective is to uncover the characteristics of I_t that are in line with the oil market. Table 6 (Panel A) shows that the average annualized $\sqrt{\mathrm{I}_t}$ is 11%, while that of $\sqrt{\mathrm{V}_t}$ is 33%, indicating that non-idiosyncratic volatility is more dominant than idiosyncratic volatility. Additionally, I observe that I_t and V_t have an imperfect correlation of 0.88. This is further demonstrated in Figure 1, which illustrates the time-series evolutions of $\sqrt{\mathrm{V}_t}$ and $\sqrt{\mathrm{I}_t}$.

I provide evidence to answer the question of how important unspanned diffusive risks are when considering the effects of idiosyncratic risks. In this regard, my estimations, based on the dynamics of non-idiosyncratic variance (i.e., $d\mathrm{V}_t = (\theta_{\mathrm{v}}^{\mathbb{P}} - \kappa_{\mathrm{v}}^{\mathbb{P}} \mathrm{V}_t)\, dt + \sigma_{\mathrm{v}}\, \rho_{\mathrm{v}} \sqrt{\mathrm{V}_t}\, d\mathbb{z}_t^{\mathbb{P}} + \sigma_{\mathrm{v}} \sqrt{1-\rho_{\mathrm{v}}^2} \sqrt{\mathrm{V}_t}\, d\mathbb{u}_t^{\mathbb{P}}$), support an estimate of $\rho_{\mathrm{v}} = -0.677$ (see Table 6 (Panel E)). This implies that a reasonable model of oil futures dynamics would likely include unspanned risks and idiosyncratic risks.

To measure the importance of idiosyncratic risks for the excess returns of options, I ask the following question: Do spanning regressions augmented with changes in idiosyncratic variance improve goodness-of-fit $\overline{R}^2$? I examine a regression specification of the type (similarly for OTM puts and calls):

$$\underbrace{\mathbb{r}^{\text{straddle}}_{\{t\to t+\Delta\}}}_{\text{excess returns of straddle}} = \text{constant} + \sum_{\ell=1}^{3} \Lambda_{\text{pc}}^{[\ell],\text{return}} \times \underbrace{\widehat{\text{PC}}^{[\ell],\text{return}}_{\{t\to t+\Delta\}}}_{\text{PC of oil futures returns}} + \Upsilon_{\text{idio}} \times \underbrace{\{\mathrm{I}_{t+\Delta} - \mathrm{I}_t\}}_{\text{model extracted}} + \underbrace{\varepsilon_{\{t\to t+\Delta\}}}_{\text{residual}}, \tag{35}$$

Table 7 (Panel A) suggests that spanning regressions augmented with changes in I_t enhance the goodness-of-fit statistics. For instance, the $\overline{R}^2$ rises from 6% to 21% for straddles. Supporting the role of idiosyncratic variance, all Υ_{idio} estimates are significantly positive, and the hypothesis of $\Upsilon_{\text{idio}} = 0$ is rejected.

Focussing further on extracted I_t, I address the relative contribution of I_t and V_t to $\mathrm{Variance}_t^{\mathbb{Q}}$ and $\mathrm{VIX}_t^{\mathrm{oil}}$. For example, I are interested in regressions of the type:

$$\underbrace{\log(\mathrm{Variance}_t^{\mathbb{Q}})}_{\text{(actual)}} = \Pi_0 + \Pi_{\mathrm{I}} \underbrace{\log(\mathrm{I}_t)}_{\text{(model extracted)}} + \Pi_{\mathrm{V}} \underbrace{\log(\mathrm{V}_t)}_{\text{(model extracted)}} + \widetilde{e}_t. \tag{36}$$

Results in Table 7 (Panel B) indicate that I_t is a determinant of movements in $\mathrm{Variance}_t^{\mathbb{Q}}$ and $\mathrm{VIX}_t^{\mathrm{oil}}$. The $\overline{R}^2$ in the univariate regression with $\log(\mathrm{I}_t)$ is 81% for $\log(\mathrm{Variance}_t^{\mathbb{Q}})$. Adding $\log(\mathrm{V}_t)$ tends to improve $\overline{R}^2$ to 95%. Additionally, the correlation between model-based $\log(\mathrm{I}_t)$ with $\log(\mathrm{Variance}_t^{\mathbb{Q}})$ is 0.90.

Turning back to Table 6, I examine the estimates of $(\kappa_{\mathrm{I}}^{\mathbb{P}}, \theta_{\mathrm{I}}^{\mathbb{P}}, \sigma_{\mathrm{I}})$, and compare them to $(\kappa_{\mathrm{v}}^{\mathbb{P}}, \theta_{\mathrm{v}}^{\mathbb{P}}, \sigma_{\mathrm{v}})$. The structure of the mean-reversion indicates that the mean-reversion of I_t is slower than that of V_t. Furthermore, my estimates are consistent with a lower long-run mean of I_t (i.e., $\frac{\theta_{\mathrm{I}}^{\mathbb{P}}}{\kappa_{\mathrm{I}}^{\mathbb{P}}}$) than for V_t. The annualized magnitude of the volatility of I_t shocks is 85.5%.

4.2. Evidence affirms oil jump risks

Statistical properties of large movements in oil futures prices can present a challenge to modeling. From this perspective, there are some aspects of my estimated model that are noteworthy. The $\lambda_{\mathrm{J}}^{\mathbb{P}}$ estimate is 0.859, implying that one could expect 10 jumps in oil prices per year. [1]

In the Merton (1976) jump specification, the parameter $\mu_{\mathrm{J}}^{\mathbb{P}}$ is related to the mean jump size. From my estimation, the $\mu_{\mathrm{J}}^{\mathbb{P}}$ estimate is 8% annualized. The estimation also suggests the absence of oil jump risk premiums, with the data not rejecting $\mu_{\mathrm{J}}^{\mathrm{rp}} = 0$ and $\lambda_{\mathrm{J}}^{\mathrm{rp}} = 0$. Lastly, I measure the contribution of jumps to spot return volatility by $\frac{\lambda_{\mathrm{J}}^{\mathbb{P}}(\{\mu_{\mathrm{J}}^{\mathbb{P}}\}^2+\{\sigma_{\mathrm{J}}^{\mathbb{P}}\}^2)}{\lambda_{\mathrm{J}}^{\mathbb{P}}(\{\mu_{\mathrm{J}}^{\mathbb{P}}\}^2+\{\sigma_{\mathrm{J}}^{\mathbb{P}}\}^2)+\mathrm{V}_t+\mathrm{I}_t}$. While this contribution varies with the state V_t and I_t, the implied mean jump contribution is 7.1%. Jumps appear to be an essential feature of oil price dynamics.

4.3. Estimated model prices agree with the panel of actual option prices

My framework differs in several aspects and I use an option pricing model — with the oil futures underlier — that incorporates stochastic idiosyncratic variance, stochastic non-idiosyncratic variance, and

[1] This estimate can be compared to the findings in Table 2, which indicates a total of 77 (87) down (up) return movements of size greater than 5% in oil futures prices over monthly option expiration cycles.

random jumps. To the best of my knowledge, the empirical implications of this model have not been tested.

In implementation, I concentrate on how the model prices 10%, 5%, and 0% OTM puts and 0%, 5%, and 10% OTM calls. Standard Fourier inversion methods give a semi-analytical representation of put and call option prices on futures, which is demonstrated below:

$$\text{put}_t[K] = \frac{F_t^{T_F}}{R_t^{\text{rf}}}\left(e^k\{\frac{1}{2}-\frac{1}{\pi}\int_0^\infty \mathfrak{Re}[\frac{e^{-i\phi k}\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi \mathbb{x}})}{i\phi}]d\phi\}-\{\frac{1}{2}-\frac{1}{\pi}\int_0^\infty \mathfrak{Re}[\frac{e^{-i\phi k}\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi \mathbb{x}+\mathbb{x}})}{i\phi}]d\phi\}\right), \quad (37)$$

$$\text{call}_t[K] = \frac{F_t^{T_F}}{R_t^{\text{rf}}}\left(\{\frac{1}{2}+\frac{1}{\pi}\int_0^\infty \mathfrak{Re}[\frac{e^{-i\phi k}\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi \mathbb{x}+\mathbb{x}})}{i\phi}]d\phi\} - e^k\{\frac{1}{2}+\frac{1}{\pi}\int_0^\infty \mathfrak{Re}[\frac{e^{-i\phi k}\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi \mathbb{x}})}{i\phi}]d\phi\}\right), \quad (38)$$

where $i=\sqrt{-1}$, $\mathfrak{Re}[.]$ is the real part of the complex-valued function, and ϕ is the integration parameter. For compactness, I write $\mathbb{x} \equiv \log(\frac{F_{t+\Delta}^{T_F}}{F_{t-}^{T_F}})$ and recall $k=\log(\frac{K}{F_t^{T_F}})$. The form of $\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi \mathbb{x}})$ (and hence $\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi \mathbb{x}+\mathbb{x}})$) — which is analytical in V_t and I_t — is shown in equation (A21) of the Appendix.

At the beginning of each oil option expiration cycle, I use the extracted V_t and I_t, and estimated parameter vector Θ, to calculate the prices of puts and calls according to equations (37) and (38). The question is whether the estimated model produces suitable levels of idiosyncratic variance and non-idiosyncratic variance to capture the cross-section of option prices by moneyness, and respects the time-series properties of option prices with fixed moneyness (I consider six such time-series).

Table 8 (Panels A through D) demonstrates the model's ability to capture option prices. I exhibit dollar pricing errors as model option price minus actual option price, while percentage pricing errors are computed as $100\times\log(\frac{\text{model price}}{\text{actual price}})$. To construct the time-series of root-mean-squared errors (RMSE) across all options, I first work out the RMSE across the six puts and calls at the beginning of the expiration cycle, and then across all option expiration cycles.

For instance, Table 8 (Panel D) reveals that the mean RMSE (average) percentage pricing error across all options is 22% (7%). The main inference from this evidence is that the model specification — with extracted idiosyncratic variance — can provide a reasonable fit to the options data.

This part of my investigation has the characteristics of out-of-sample validation, as the time-series of

individual put and call prices were not included in the measurement equations. My analysis shows that error metrics for $100 \times \log(\frac{\text{model price}}{\text{actual price}})$ are comparable across puts and calls (Table 8 (Panel B)). The dollar pricing errors (Table 8 (Panel A)) give another viewpoint on model adequacy. In comparison to the average ATM put (call) price of \$1.84 (\$1.84) in the data, the mean absolute pricing error is \$0.22 (\$0.11).

To further probe the option model, I look at the normalized option prices as the dependent variable, and then run regressions to analyze the data, as follows:

$$\log(\frac{\text{actual option price}_t}{\text{futures price}_t}) \;=\; \alpha + \beta \log(\frac{\text{model option price}_t}{\text{futures price}_t}) \;+\; \widetilde{e}_t. \tag{39}$$

Table 8 (Panel C) shows that the β coefficients are between 0.89 and 0.97, and $\overline{R}^2$ values between 83% and 89%, which indicates evidence of model conformity. Additionally, Table 5 reveals the percentile values of modeled and actual prices are all in agreement. The evidence indicates that the option model is consistent with the data.

I also investigate inference by running a regression to analyze the percentage pricing errors in relation to three principal components of oil futures returns and two principal components of oil futures return variance. The variance principal components were calculated from monthly variances (using daily returns of futures with maturity of 1-, 3-, 6-, 9-, and 12-month). The specification for this regression is as follows:

$$\underbrace{\log(\frac{\text{model price}_t}{\text{actual price}_t})}_{\text{percentage option pricing error}} \;=\; \text{constant} \;+\; \sum_{\ell=1}^{3} \mathbb{b}_{\text{pc}}^{[\ell],\text{return}} \times \underbrace{\widehat{\text{PC}}_{\{t-1\to t\}}^{[\ell],\text{return}}}_{\text{PC of oil futures returns}} \;+\; \sum_{\ell=1}^{2} \mathbb{b}_{\text{pc}}^{[\ell],\text{variance}} \times \underbrace{\widehat{\text{PC}}_{\{t-1\to t\}}^{[\ell],\text{variance}}}_{\text{PC of return variance}} \;+\; \underbrace{\varepsilon_t,}_{\text{residual}}$$

Table 8 (Panel E) shows that lagged information does not explain variation in pricing errors, as indicated by the low $\overline{R}^2$ values of 3% to 6%. Overall, the option model with V_t and I_t exhibits reasonable performance.

4.4. Model tracks the behavior of $\sqrt{\text{Variance}_t^{\mathbb{Q}}}$ *and* $\text{VIX}_t^{\text{oil}}$

Idiosyncratic variance and jumps are substantive facts of the oil market and I account for them in model development. This resulted in predictions for the time-series of option-inferred $\sqrt{\text{Variance}_t^{\mathbb{Q}}}$, and $\text{VIX}_t^{\text{oil}}$. Options are useful for this purpose, as they provide the means to calculate the risk-neutral expectations of power payoffs underlying the oil futures return distribution at each t.

In my investigation, I analyze the differences between the model-implied risk-neutral moments and

their actual counterparts. Additionally, I use the following type of regressions to assess the model implications for data moments:

$$\underbrace{\log(\sqrt{\text{Variance}_t^{\mathbb{Q}}})}_{\text{(actual)}} = \alpha + \beta \underbrace{\log(\sqrt{\text{Variance}_t^{\mathbb{Q}}})}_{\text{(model implied)}} + \widetilde{e}_t. \tag{40}$$

I acknowledge that the accuracy of my conclusions relies on the precision of the measurements of these quantities, and I pay special attention to the number of OTM puts and calls used in the construction. The model's adequacy is judged by analyzing the deviations of α and β from zero and one, respectively. In the model calculations, I use the extracted V_t and I_t.

Table 9 reveals that the model is providing a good match for $\sqrt{\text{Variance}_t^{\mathbb{Q}}}$ and $\text{VIX}_t^{\text{oil}}$. For instance, the RMSE is 0.0% and 1.7%, and the corresponding α and β values are close to zero and one, respectively. Figure 2 illustrates the model fit for $\text{VIX}_t^{\text{oil}}$. A rejection of $\beta = 0$ indicates that the modeled quantity contains information about the actual. Additionally, the model is also providing a good fit to the log of the futures price.

4.5. Estimated model can produce negative risk premiums of OTM calls on oil futures

This part of my paper seeks to answer the question of whether the estimated model is aligned with the expectation of state contingent futures returns under both the real-world and the risk-neutral probability measures? To address this, I note that the risk premium of an OTM call option on futures satisfies

$$\underbrace{\text{Risk Premium}^{\text{call}}_{\{t\to t+\Delta\}}}_{\text{expected excess return}} \equiv \frac{\mathbb{E}_t^{\mathbb{P}}(\max(F_{t+\Delta}^{T_F} - K, 0))}{\frac{1}{R^{\text{rf}}_{\{t\to t+\Delta\}}}\mathbb{E}_t^{\mathbb{Q}}(\max(F_{t+\Delta}^{T_F} - K, 0))} - R^{\text{rf}}_{\{t\to t+\Delta\}}) \tag{41}$$

$$= R^{\text{rf}}_{\{t\to t+\Delta\}}(\frac{\mathbb{E}_t^{\mathbb{P}}(\max(e^{\mathbb{x}} - e^k, 0))}{\mathbb{E}_t^{\mathbb{Q}}(\max(e^{\mathbb{x}} - e^k, 0))} - 1), \tag{42}$$

where $\mathbb{x} \equiv \log(\frac{F_{t+\Delta}^{T_F}}{F_{t-}^{T_F}})$ and $k = \log(\frac{K}{F_t^{T_F}})$. I derive as follows:

$$\frac{\mathbb{E}_t^{\mathbb{P}}(\max(e^{\mathbb{x}} - e^k, 0))}{F_t^{T_F}} = \mathbb{E}_t^{\mathbb{P}}(e^{\mathbb{x}})\{\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathfrak{Re}[\frac{e^{-i\phi k}\frac{\mathbb{E}_t^{\mathbb{P}}(e^{(i\phi+1)\mathbb{x}})}{\mathbb{E}_t^{\mathbb{P}}(e^{\mathbb{x}})}}{i\phi}]d\phi\} - e^k\{\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathfrak{Re}[e^{-i\phi k}\frac{\mathbb{E}_t^{\mathbb{P}}(e^{i\phi\mathbb{x}})}{i\phi}]d\phi\}, \tag{43}$$

and

$$\frac{\mathbb{E}_t^{\mathbb{Q}}(\max(e^{\mathbb{x}} - e^k, 0))}{F_t^{T_F}} = \{\frac{1}{2} + \frac{1}{\pi}\int_0^{\infty} \mathfrak{Re}[\frac{e^{-i\phi k}\,\mathbb{E}_t^{\mathbb{Q}}(e^{(i\phi+1)\mathbb{x}})}{i\phi}]d\phi\} - e^k\{\frac{1}{2} + \frac{1}{\pi}\int_0^{\infty} \mathfrak{Re}[e^{-i\phi k}\,\frac{\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi\mathbb{x}})}{i\phi}]d\phi\}. \tag{44}$$

I present the parametric forms of $\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi\mathbb{x}})$ and $\mathbb{E}_t^{\mathbb{P}}(e^{i\phi\mathbb{x}})$ in equations (A21) and (A46) of the Appendix. The solution for the call option risk premium is noteworthy due to the specific properties of the pricing kernel in (7), as well as the futures price dynamics with idiosyncratic, jump, unspanned, and spanned risks in (19). The semi-analytical expression for put option risk premium follows from taking $\mathbb{P}$ and $\mathbb{Q}$ expectation of $\max(K - F_{t+\Delta}^{T_F}, 0) = \max(F_{t+\Delta}^{T_F} - K, 0) - F_{t+\Delta}^{T_F} + K$ and rearranging.

Table 10 shows the degree to which the model can generate negative expected excess returns of OTM call options. To calculate risk premiums of OTM calls, I use the estimated parameters, V_t, and I_t at the beginning of the oil option expiration cycles. I find that the mean risk premium of 10% OTM call options is -14%, which is within the bootstrap confidence intervals of average option return of $[-66\%, -11\%]$ from the data. Similarly, the mean risk premium of 10% OTM put options is -16% from the model, which is within the bootstrap confidence intervals of $[-53\%, 6\%]$ from the data. The mean straddle risk premium is -1%, and also in line with the data on excess return of straddles.

My estimations identify the sources of model risk premiums. For instance, the estimated χ_{u}^* is negative and statistically significant, whereas χ_{v}^* is reliably positive. This implies that $\kappa_{\mathrm{v}}^{\mathbb{Q}} < \kappa_{\mathrm{v}}^{\mathbb{P}}$ in equation (16), since $\chi_{\mathrm{v}}^* \sigma_{\mathrm{v}} \rho_{\mathrm{v}} < 0$, $\chi_{\mathrm{u}}^* \sigma_{\mathrm{v}} \sqrt{1-\rho_{\mathrm{v}}^2} < 0$, and $\rho_{\mathrm{v}} < 0$. This estimated pricing kernel explains the risk premiums of OTM puts, straddles, and OTM calls, while providing a reasonable fitting errors for option prices, realized volatilities, and options-inferred $\mathrm{VIX}_t^{\mathrm{oil}}$.

5. Conclusion

I present an approach that models both idiosyncratic risks and jump risks for crude oil. My results reveal that models that incorporate both of these risks are more aligned with multiple aspects of the crude oil market. My empirical evidence characterizing the data shows that the variation in excess option returns is linked to idiosyncratic risks and/or unspanned variance risks. The latter risks are distinct in that they are included in the pricing kernel. My evidence reinforces the mechanism of oil idiosyncratic risks, with $\mathrm{VIX}_t^{\mathrm{oil}}$

being substantially higher than $\text{VIX}_t^{\text{S\&P500}}$, the option-inferred risk-neutral return skewness changing signs, and risk-neutral return kurtosis being elevated. Furthermore, the average excess returns associated with buying deep out-of-the-money calls on crude oil futures is significantly negative. This evidence supports the view that increasing oil prices are a cause for concern among oil market participants.[2]

This study makes a contribution by investigating models that differentiate between idiosyncratic risks, jump risks, unspanned risks, and spanned risks. My model considers two state-variables, namely, idiosyncratic variance and non-idiosyncratic variance, and uses them to derive analytical solutions for $\text{VIX}_t^{\text{oil}}$ and risk-neutral return moments, as well as semi-analytical solutions for option risk premiums. The benefit of my model representations is that they enable us to assess the alignment of empirical observations in the market for crude oil options and futures.

I use Kalman filtering and maximum-likelihood estimation to extract the time-series of idiosyncratic variances and non-idiosyncratic variances. The models are estimated using data on crude oil options, return volatilities, and futures prices from January 1990 to December 2019. My results demonstrate that (i) oil idiosyncratic variance is empirically relevant, (ii) the estimation identifies jump risks, and (iii) idiosyncratic risks are influential in option pricing and model performance. The estimated model matches variations in risk-neutral return moments and is consistent with the negative sign of OTM call option risk premiums.

[2]Much has been explored about oil futures risk premiums but not about state-contingent futures risk premiums. My approach to study state-contingent futures risk premiums — using option excess returns — departs from Bessembinder and Chan (1992), Trolle and Schwartz (2009, 2010), Hong and Yogo (2012), Casassus, Liu, and Tang (2013), Szymanowska, De Roon, Nijman, and Van den Goorbergh (2014), Christoffersen, Jacobs, and Li (2016), Ready, Roussanov, and Ward (2017), and Gao, Hitzemann, Shaliastovich, and Xu (2022).

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Appendix

A. Expressions employed in the paper

Expressions presented here are a consequence of solving known Ricatti equations. Hence, I omit detailed steps. For brevity, $\tau \equiv T_F - t$.

Oil futures prices. Writing the futures price as $\frac{F_t^{T_F}}{S_t} = \exp(\mathbb{a}_{\text{fut}}[\tau] + \mathbb{b}_{\text{fut}}[\tau]\mathrm{V}_t + \mathbb{c}_{\text{fut}}[\tau]\mathrm{I}_t)$, I determine that

$$\mathbb{a}_{\text{fut}}[\tau] = \delta_{\mathbf{y}}^{\circ}\tau + \mathbb{a}_{\mathrm{v}}[\tau] + \mathbb{a}_{\mathrm{I}}[\tau], \tag{A1}$$

$$\mathbb{b}_{\text{fut}}[\tau] = \frac{2(A+\frac{1}{2}B^2)(1-e^{-\psi_{\mathrm{v}}\tau})}{2\psi_{\mathrm{v}} - (\psi_{\mathrm{v}} - (\kappa_{\mathrm{v}}^{\mathbb{Q}} - \sigma_{\mathrm{v}}B))(1-e^{-\psi_{\mathrm{v}}\tau})}, \tag{A2}$$

$$\mathbb{a}_{\mathrm{v}}[\tau] = -\frac{\theta_{\mathrm{v}}^{\mathbb{Q}}}{\sigma_{\mathrm{v}}^2}\left[(\psi_{\mathrm{v}} - (\kappa_{\mathrm{v}}^{\mathbb{Q}} - \sigma_{\mathrm{v}}B))\tau + 2\log(1 - \frac{(\psi_{\mathrm{v}} - (\kappa_{\mathrm{v}}^{\mathbb{Q}} - \sigma_{\mathrm{v}}B))(1-e^{-\psi_{\mathrm{v}}\tau})}{2\psi_{\mathrm{v}}})\right], \tag{A3}$$

$$\psi_{\mathrm{v}} = \sqrt{(\kappa_{\mathrm{v}}^{\mathbb{Q}} - \sigma_{\mathrm{v}}B)^2 - 2\sigma_{\mathrm{v}}^2(A + \frac{1}{2}B^2)}, \quad \text{and} \tag{A4}$$

$$A = \delta_{\mathrm{v}}^{*} \qquad \text{and} \qquad B = 0. \tag{A5}$$

Noting that $\kappa_{\mathrm{I}}^{\mathbb{Q}} = \kappa_{\mathrm{I}}^{\mathbb{P}}$ and $\theta_{\mathrm{I}}^{\mathbb{Q}} = \theta_{\mathrm{I}}^{\mathbb{P}}$, I also obtain the following:

$$\mathbb{c}_{\text{fut}}[\tau] = \frac{2(A+\frac{1}{2}B^2)(1-e^{-\psi_{\mathrm{I}}\tau})}{2\psi_{\mathrm{I}} - (\psi_{\mathrm{I}} - (\kappa_{\mathrm{I}}^{\mathbb{Q}} - \sigma_{\mathrm{I}}B))(1-e^{-\psi_{\mathrm{I}}\tau})}, \tag{A6}$$

$$\mathbb{a}_{\mathrm{I}}[\tau] = -\frac{\theta_{\mathrm{I}}^{\mathbb{Q}}}{\sigma_{\mathrm{I}}^2}\left[(\psi_{\mathrm{I}} - (\kappa_{\mathrm{I}}^{\mathbb{Q}} - \sigma_{\mathrm{I}}B))\tau + 2\log(1 - \frac{(\psi_{\mathrm{I}} - (\kappa_{\mathrm{I}}^{\mathbb{Q}} - \sigma_{\mathrm{I}}B))(1-e^{-\psi_{\mathrm{I}}\tau})}{2\psi_{\mathrm{I}}})\right], \tag{A7}$$

$$\psi_{\mathrm{I}} = \sqrt{(\kappa_{\mathrm{I}}^{\mathbb{Q}} - \sigma_{\mathrm{I}}B)^2 - 2\sigma_{\mathrm{I}}^2(A + \frac{1}{2}B^2)}, \quad \text{and} \tag{A8}$$

$$A = \delta_{\mathrm{I}}^{*} \qquad \text{and} \qquad B = 0. \tag{A9}$$

Ito's lemma applied to $F_t^{T_F} = S_t \exp(\mathbb{a}_{\text{fut}}[\tau] + \mathbb{b}_{\text{fut}}[\tau]\mathrm{V}_t + \mathbb{c}_{\text{fut}}[\tau]\mathrm{I}_t)$ implies the following dynamics under $\mathbb{P}$:

$$\frac{dF_t^{T_F}}{F_t^{T_F}} = \frac{dS_t}{S_t} + \mathbb{b}_{\text{fut}}[\tau]\,d\mathrm{V}_t + \mathbb{c}_{\text{fut}}[\tau]d\mathrm{I}_t + \text{cross-products} \tag{A10}$$

$$= (\mathbf{y}_t - \frac{1}{dt}\mathrm{cov}_t^{\mathbb{P}}(\frac{dM_t}{M_t}, \frac{dS_t}{S_t}))\,dt + \sqrt{\mathrm{V}_t}\,d\mathbb{z}_t^{\mathbb{P}} + \sqrt{\mathrm{I}_t}\,d\varepsilon_t^{\mathbb{P}} + \{(e^{J_t}-1)d\mathbb{N}_t^{\mathbb{P}} - \lambda_{\mathbf{J}}^{\mathbb{P}}\,\mathbb{E}_t^{\mathbb{P}}(e^{J_t}-1)dt\}$$

$$+ \mathbb{b}_{\text{fut}}[\tau]\{\sigma_{\mathrm{v}}\rho_{\mathrm{v}}\sqrt{\mathrm{V}_t}\,d\mathbb{z}_t^{\mathbb{P}} + \sigma_{\mathrm{v}}\sqrt{1-\rho_{\mathrm{v}}^2}\sqrt{\mathrm{V}_t}\,d\mathbb{u}_t^{\mathbb{P}}\} + \mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\mathrm{I}}\sqrt{\mathrm{I}_t}\,d\varepsilon_t^{\mathbb{P}}. \tag{A11}$$

Likewise, I have under $\mathbb{Q}$ that

$$
\begin{aligned}
d\log(F_t^{T_F}) &= \{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\rho_{\text{v}}\}\sqrt{\text{V}_t}\,d\mathbb{z}_t^{\mathbb{Q}} \;+\; \mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\sqrt{\text{V}_t}\,d\mathbb{u}_t^{\mathbb{Q}} \\
&+ \{1+\mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\text{I}}\}\sqrt{\text{I}_t}d\varepsilon_t^{\mathbb{Q}} \\
&- \frac{1}{2}\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\rho_{\text{v}}\}^2\text{V}_t - \frac{1}{2}\{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2\,\text{V}_t - \frac{1}{2}\{1+\mathbb{c}_{\text{fut}}[\tau]\,\sigma_{\text{I}}\}^2\,\text{I}_t \\
&+ J_t\,d\mathbb{N}_t^{\mathbb{Q}} \;-\; \lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1)dt. \qquad \text{(A12)}
\end{aligned}
$$

I use equation (A12) to derive the characteristic function of $\log(F_{t+\Delta}^{T_F})$ under $\mathbb{Q}$. ■

Oil VIX. The expression for $\text{VIX}_t^{\text{oil}}$ is $\{\text{VIX}_t^{\text{oil}}\}^2 = \mathbb{a}_{\text{vix}}[\Delta] \;+\; \mathbb{b}_{\text{vix}}[\Delta]\times\text{V}_t \;+\; \mathbb{c}_{\text{vix}}[\Delta]\times\text{I}_t$, where

$$
\begin{aligned}
\mathbb{a}_{\text{vix}}[\Delta] &= 2\lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1-J_t) \\
&+\frac{1}{\Delta}(\{1+\sigma_{\text{v}}\rho_{\text{v}}\,\mathbb{b}_{\text{fut}}[\tau]\}^2+\{\sigma_{\text{v}}\,\mathbb{b}_{\text{fut}}[\tau]\sqrt{1-\rho_{\text{v}}^2}\}^2)[\frac{\Delta\theta_{\text{v}}^{\mathbb{Q}}}{\kappa_{\text{v}}^{\mathbb{Q}}}+\frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{Q}}\Delta})}{\kappa_{\text{v}}^{\mathbb{Q}}}\{-\frac{\theta_{\text{v}}^{\mathbb{Q}}}{\kappa_{\text{v}}^{\mathbb{Q}}}\}] \\
&+\frac{1}{\Delta}\{1+\sigma_{\text{I}}\,\mathbb{c}_{\text{fut}}[\tau]\}^2[\frac{\Delta\theta_{\text{I}}^{\mathbb{P}}}{\kappa_{\text{I}}^{\mathbb{P}}}+\frac{(1-e^{-\kappa_{\text{I}}^{\mathbb{P}}\Delta})}{\kappa_{\text{I}}^{\mathbb{P}}}\{-\frac{\theta_{\text{I}}^{\mathbb{P}}}{\kappa_{\text{I}}^{\mathbb{P}}}\}], \qquad \text{(A13)} \\
\mathbb{b}_{\text{vix}}[\Delta] &= \frac{1}{\Delta}(\{1+\sigma_{\text{v}}\rho_{\text{v}}\,\mathbb{b}_{\text{fut}}[\tau]\}^2+\{\sigma_{\text{v}}\,\mathbb{b}_{\text{fut}}[\tau]\sqrt{1-\rho_{\text{v}}^2}\}^2)[\frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{Q}}\Delta})}{\kappa_{\text{v}}^{\mathbb{Q}}}], \qquad \text{(A14)} \\
\mathbb{c}_{\text{vix}}[\Delta] &= \frac{1}{\Delta}\{1+\sigma_{\text{I}}\,\mathbb{c}_{\text{fut}}[\tau]\}^2[\frac{(1-e^{-\kappa_{\text{I}}^{\mathbb{P}}\Delta})}{\kappa_{\text{I}}^{\mathbb{P}}}], \qquad \text{(A15)} \\
\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1-J_t) &= \exp(\mu_{\text{J}}^{\mathbb{Q}}+\frac{1}{2}\{\sigma_{\text{J}}^{\mathbb{Q}}\}^2)-1-\mu_{\text{J}}^{\mathbb{Q}}, \qquad \text{(A16)} \\
\lambda_{\text{J}}^{\mathbb{Q}} &= \lambda_{\text{J}}^{\mathbb{P}}+\{\lambda_{\text{J}}^{\text{rp}}\}^2 \;\text{ and }\; \mu_{\text{J}}^{\mathbb{Q}} \;=\; \mu_{\text{J}}^{\mathbb{P}}\times\exp(\mu_{\text{J}}^{\text{rp}}) \;\text{ and }\; \{\sigma_{\text{J}}^{\mathbb{Q}}\}^2=\{\sigma_{\text{J}}^{\mathbb{P}}\}^2\times\exp(\sigma_{\text{J}}^{\text{rp}}). \qquad \text{(A17)}
\end{aligned}
$$

This specification accommodates

$$
\lambda_{\text{J}}^{\mathbb{Q}} > \lambda_{\text{J}}^{\mathbb{P}}, \qquad |\mu_{\text{J}}^{\mathbb{Q}}| > |\mu_{\text{J}}^{\mathbb{P}}| \qquad \text{and} \qquad \sigma_{\text{J}}^{\mathbb{Q}} > \sigma_{\text{J}}^{\mathbb{P}}. \qquad \text{(A18)}
$$

Throughout, I use the cumulant generating function of $\sum_{\ell=1}^{\mathbb{N}_{t+\Delta}^{\mathbb{Q}}} J_\ell$ under $\mathbb{P}$ and $\mathbb{Q}$, as follows:

$$
\text{cumulant}^{\mathbb{Q}}[\phi] = \log\mathbb{E}_t^{\mathbb{Q}}(\exp(\phi\sum_{\ell=1}^{\mathbb{N}_{t+\Delta}^{\mathbb{Q}}} J_\ell)) = \lambda_{\text{J}}^{\mathbb{Q}}\,\Delta\{e^{\phi\mu_{\text{J}}^{\mathbb{Q}}+\frac{1}{2}\phi^2\{\sigma_{\text{J}}^{\mathbb{Q}}\}^2} \;-\; 1\}. \qquad \text{(A19)}
$$

I obtain the central moments through the cumulants. ■

Characteristic function under $\mathbb{Q}$. For $i=\sqrt{-1}$, I need conditional expectation of the type

$$\begin{aligned}
\mathbb{E}_t^{\mathbb{Q}}(\exp(i\phi\log\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}})) &= \mathbb{E}_t^{\mathbb{Q}}(\exp\Big(i\phi\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\rho_{\text{v}}\}\int_t^{t+\Delta}\sqrt{\text{V}_\ell}\,d\mathbb{z}_\ell^{\mathbb{Q}} \;+\; i\phi\{\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}\int_t^{t+\Delta}\sqrt{\text{V}_\ell}\,d\mathbb{u}_\ell^{\mathbb{Q}} \\
&\quad + i\phi\{1+\mathbb{c}_{\text{fut}}[\tau]\sigma_{\text{I}}\}\int_t^{t+\Delta}\sqrt{\text{I}_\ell}\,d\varepsilon_\ell^{\mathbb{Q}} \;+\; \{\;-\;i\phi\lambda_{\text{J}}^{\mathbb{Q}}\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1)\Delta\} \\
&\quad -[\frac{i\phi}{2}\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\rho_{\text{v}}\}^2+\frac{i\phi}{2}\{\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2]\int_t^{t+\Delta}\text{V}_\ell\,d\ell \\
&\quad -\frac{i\phi}{2}\{1+\mathbb{c}_{\text{fut}}[\tau]\sigma_{\text{I}}\}^2\int_t^{t+\Delta}\text{I}_\ell\,d\ell \;+\; \sum_{\ell=1}^{\mathbb{N}_{t+\Delta}^{\mathbb{Q}}} i\phi J_\ell\Big). \qquad \text{(A20)}
\end{aligned}$$

I determine that

$$\mathbb{H}_t^{\mathbb{Q}}[\phi] \;\equiv\; \mathbb{E}_t^{\mathbb{Q}}(e^{i\phi\log(\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}})}) \;=\; \exp\left(\mathbb{a}_{\text{cf}}[\phi]+\mathbb{b}_{\text{cf}}[\phi]\times\text{V}_t+\mathbb{c}_{\text{cf}}[\phi]\times\text{I}_t\right), \qquad \text{(A21)}$$

where

$$\begin{aligned}
\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi J_t}) &= \exp(i\phi\mu_{\text{J}}^{\mathbb{Q}}+\frac{1}{2}\{i\phi\}^2\{\sigma_{\text{J}}^{\mathbb{Q}}\}^2), && \text{(A22)}\\
\mathbb{a}_{\text{J}}[\phi] &= -\,i\phi\Delta\lambda_{\text{J}}^{\mathbb{Q}}\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1)+\lambda_{\text{J}}^{\mathbb{Q}}\Delta\{\mathbb{E}_t^{\mathbb{Q}}(e^{i\phi J_t})\;-\;1\}, && \text{(A23)}\\
\mathbb{a}_{\text{cf}}[\phi] &= \mathbb{a}_{\text{J}}[\Delta;\phi]+\mathbb{a}_{\text{v}}[\Delta;\phi]+\mathbb{a}_{\text{I}}[\Delta;\phi], \quad \mathbb{b}_{\text{cf}}[\phi] \;=\; \mathbb{b}_{\text{v}}[\Delta;\phi] \quad \text{and} \quad \mathbb{c}_{\text{cf}}[\phi] \;=\; \mathbb{b}_{\text{I}}[\Delta;\phi]. && \text{(A24)}
\end{aligned}$$

The solution is (the dependence on ϕ is suppressed for brevity) as follows:

$$\begin{aligned}
\alpha^* &= i\phi\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\rho_{\text{v}}\}, && \text{(A25)}\\
\beta^* &= i\phi\{\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}, && \text{(A26)}\\
a^* &= \kappa_{\text{v}}^{\mathbb{Q}}-\sigma_{\text{v}}\rho_{\text{v}}\alpha^*-\beta^*\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}, && \text{(A27)}\\
A &= -\frac{i\phi}{2}[\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\rho_{\text{v}}\}^2+\{\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2], && \text{(A28)}\\
B^2 &= \{\alpha^*\}^2+\{\beta^*\}^2, && \text{(A29)}\\
\psi_{\text{v}} &= \sqrt{\{a^*\}^2-2\sigma_{\text{v}}^2(A+\frac{1}{2}B^2)}, && \text{(A30)}\\
\mathbb{b}_{\text{v}}[\Delta] &= \frac{2(A+\frac{1}{2}B^2)(1-e^{-\psi_{\text{v}}\tau})}{2\psi_{\text{v}}-(\psi_{\text{v}}-\{a^*\})(1-e^{-\psi_{\text{v}}\Delta})}, \quad \text{and} \\
\mathbb{a}_{\text{v}}[\Delta] &= -\frac{\theta_{\text{v}}^{\mathbb{Q}}}{\sigma_{\text{v}}^2}\left[(\psi_{\text{v}}-\{a^*\})\Delta+2\log(1-\frac{(\psi_{\text{v}}-\{a^*\})(1-e^{-\psi_{\text{v}}\Delta})}{2\psi_{\text{v}}})\right]. && \text{(A31)}
\end{aligned}$$

Furthermore,

$$
\begin{aligned}
\mathbb{b}_{\mathrm{I}}[\Delta] &= \frac{2(A+\frac{1}{2}B^2)(1-e^{-\psi_{\mathrm{I}}\Delta})}{2\psi_{\mathrm{I}}-(\psi_{\mathrm{I}}-(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B))(1-e^{-\psi_{\mathrm{I}}\Delta})}, \\
\mathbb{a}_{\mathrm{I}}[\Delta] &= -\frac{\theta_{\mathrm{I}}^{\mathbb{Q}}}{\sigma_{\mathrm{I}}^2}\left[(\psi_{\mathrm{I}}-(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B))\Delta+2\log(1-\frac{(\psi_{\mathrm{I}}-(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B))(1-e^{-\psi_{\mathrm{I}}\Delta})}{2\psi_{\mathrm{I}}})\right], \qquad \text{(A32)} \\
\psi_{\mathrm{I}} &= \sqrt{(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B)^2-2\sigma_{\mathrm{I}}^2(A+\frac{1}{2}B^2)}, \qquad \text{(A33)} \\
A &= -\frac{i\phi}{2}\{1+\mathbb{c}_{\mathrm{fut}}[\tau]\,\sigma_{\mathrm{I}}\}^2 \quad \text{and} \quad B = i\phi\{1+\sigma_{\mathrm{I}}\,\mathbb{c}_{\mathrm{fut}}[\tau]\}. \qquad \text{(A34)}
\end{aligned}
$$

This expression is the build block for determining central moments and option prices. ■

The variance under Q. The price of volatility payoff is

$$
\begin{aligned}
\underbrace{\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}-1\}^2)}_{\text{futures return variance}} &= \mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}\}^2)+1-2\,\underbrace{\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}\})}_{=1} = \mathbb{H}_t^{\mathbb{Q}}[\phi]\Big|_{\phi=\frac{2}{i}} - 1 \qquad \text{(A35)} \\
&= e^{\mathbb{a}_{\mathrm{vol}}[\Delta]\,+\,\mathbb{b}_{\mathrm{vol}}[\Delta]\times \mathrm{V}_t\,+\,\mathbb{c}_{\mathrm{vol}}[\Delta]\times \mathrm{I}_t} - 1, \qquad \text{(A36)}
\end{aligned}
$$

where,

$$
\mathbb{a}_{\mathrm{vol}}[\Delta] = \mathbb{a}_{\mathrm{cf}}[\phi]\Big|_{\phi=\frac{2}{i}}, \qquad \mathbb{b}_{\mathrm{vol}}[\Delta] = \mathbb{b}_{\mathrm{cf}}[\phi]\Big|_{\phi=\frac{2}{i}} \qquad \text{and} \qquad \mathbb{c}_{\mathrm{vol}}[\Delta] = \mathbb{c}_{\mathrm{cf}}[\phi]\Big|_{\phi=\frac{2}{i}}. \qquad \text{(A37)}
$$

I have the expressions for Result 3. ■

Expected integrated variance of futures returns under $\mathbb{P}$. Based on (19), the instantaneous variance of log futures return is

$$
\begin{aligned}
\mathrm{var}_t^{\mathbb{P}}[d\log(F_t^{T_F})]/dt &= (\{1+\mathbb{b}_{\mathrm{fut}}[\tau]\,\sigma_{\mathrm{v}}\,\rho_{\mathrm{v}}\}^2 + \{\mathbb{b}_{\mathrm{fut}}[\tau]\,\sigma_{\mathrm{v}}\sqrt{1-\rho_{\mathrm{v}}^2}\}^2)\,\mathrm{V}_t \\
&+ \{1+\mathbb{c}_{\mathrm{fut}}[\tau]\,\sigma_{\mathrm{I}}\}^2\,\mathrm{I}_t + \lambda_{\mathrm{J}}^{\mathbb{P}}(\{\mu_{\mathrm{J}}^{\mathbb{P}}\}^2+\{\sigma_{\mathrm{J}}^{\mathbb{P}}\}^2).
\end{aligned}
$$

Hence, I obtain $\mathrm{rv}_{\{t\to t+\Delta\}}^{\mathrm{futures},\mathbb{P}} = \mathbb{a}_{\mathrm{rv}}^{\mathrm{fut}}[\Delta]+\mathbb{b}_{\mathrm{rv}}^{\mathrm{fut}}[\Delta]\times \mathrm{V}_t+\mathbb{c}_{\mathrm{rv}}^{\mathrm{fut}}[\Delta]\times \mathrm{I}_t$. Finally,

$$
\begin{aligned}
\mathbb{a}_{\text{rv}}^{\text{fut}}[\Delta] &= (\{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}^2 \;+\; \{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2)\{\frac{\Delta\,\theta_{\text{v}}^{\mathbb{P}}}{\kappa_{\text{v}}^{\mathbb{P}}} + \frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{v}}^{\mathbb{P}}}\{-\frac{\theta_{\text{v}}^{\mathbb{P}}}{\kappa_{\text{v}}^{\mathbb{P}}}\}\} \\
&\quad +\{1+\sigma_{\text{I}}\,\mathbb{c}_{\text{fut}}[\tau]\}^2\{\frac{\Delta\,\theta_{\text{I}}^{\mathbb{P}}}{\kappa_{\text{I}}^{\mathbb{P}}} + \frac{(1-e^{-\kappa_{\text{I}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{I}}^{\mathbb{P}}}\{-\frac{\theta_{\text{I}}^{\mathbb{P}}}{\kappa_{\text{I}}^{\mathbb{P}}}\}\} \;+\; \Delta\times\lambda_{\text{J}}^{\mathbb{P}}(\{\mu_{\text{J}}^{\mathbb{P}}\}^2 \;+\; \{\sigma_{\text{J}}^{\mathbb{P}}\}^2), \qquad \text{(A38)} \\
\mathbb{b}_{\text{rv}}^{\text{fut}}[\Delta] &= (\{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}^2 \;+\; \{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2)\{\frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{v}}^{\mathbb{P}}}\}, \quad \text{and} \qquad \text{(A39)} \\
\mathbb{c}_{\text{rv}}^{\text{fut}}[\Delta] &= \{1+\sigma_{\text{I}}\,\mathbb{c}_{\text{fut}}[\tau]\}^2\{\frac{(1-e^{-\kappa_{\text{I}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{I}}^{\mathbb{P}}}\}. \qquad \text{(A40)}
\end{aligned}
$$

I have shown the first part of Result 4. ■

Expected integrated variance of spot returns under $\mathbb{P}$. The instantaneous variance of spot returns under $\mathbb{P}$ has the following form: $\mathbb{v}_t^{\text{spot}} \equiv \frac{1}{dt}\text{var}_t^{\mathbb{P}}[d\log(S_t)] = \lambda_{\text{J}}^{\mathbb{P}}(\{\mu_{\text{J}}^{\mathbb{P}}\}^2 + \{\sigma_{\text{J}}^{\mathbb{P}}\}^2) + \text{V}_t + \text{I}_t$. Define the expected integrated spot variance over a month as $\text{rv}_{\{t\to t+\Delta\}}^{\text{spot},\mathbb{P}} \equiv \mathbb{E}_t^{\mathbb{P}}(\int_t^{t+\Delta}\mathbb{v}_\ell^{\text{spot}}\,d\ell)$. I obtain $\text{rv}_{\{t\to t+\Delta\}}^{\text{spot},\mathbb{P}} = \mathbb{a}_{\text{rv}}^{\text{spot}}[\Delta] \;+\; \mathbb{b}_{\text{rv}}^{\text{spot}}[\Delta]\,\text{V}_t \;+\; \mathbb{c}_{\text{rv}}^{\text{spot}}[\Delta]\,\text{I}_t$, where

$$
\begin{aligned}
\mathbb{a}_{\text{rv}}^{\text{spot}}[\Delta] &= \{\frac{\Delta\,\theta_{\text{v}}^{\mathbb{P}}}{\kappa_{\text{v}}^{\mathbb{P}}} + \frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{v}}^{\mathbb{P}}}\{-\frac{\theta_{\text{v}}^{\mathbb{P}}}{\kappa_{\text{v}}^{\mathbb{P}}}\}\} + \{\frac{\Delta\,\theta_{\text{I}}^{\mathbb{P}}}{\kappa_{\text{I}}^{\mathbb{P}}} + \frac{(1-e^{-\kappa_{\text{I}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{I}}^{\mathbb{P}}}\{-\frac{\theta_{\text{I}}^{\mathbb{P}}}{\kappa_{\text{I}}^{\mathbb{P}}}\}\} \\
&\quad +\Delta\times\lambda_{\text{J}}^{\mathbb{P}}(\{\mu_{\text{J}}^{\mathbb{P}}\}^2 \;+\; \{\sigma_{\text{J}}^{\mathbb{P}}\}^2), \qquad \text{(A41)} \\
\mathbb{b}_{\text{rv}}^{\text{spot}}[\Delta] &= \frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{v}}^{\mathbb{P}}} \quad \text{and} \quad \mathbb{c}_{\text{rv}}^{\text{spot}}[\Delta] \;=\; \frac{(1-e^{-\kappa_{\text{I}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{I}}^{\mathbb{P}}}. \qquad \text{(A42)}
\end{aligned}
$$

I have shown the second part of Result 4. ■

Variance spreads. I determine that

$$
\text{VSPREAD}_{\{t\to t+\Delta\}} = \underbrace{\mathbb{a}_{\text{rv}}^{\text{fut}}[\Delta] - \Delta\times\mathbb{a}_{\text{vix}}[\Delta]}_{\equiv\mathbb{a}_{\text{spread}}[\Delta]} + \underbrace{(\mathbb{b}_{\text{rv}}^{\text{fut}}[\Delta] - \Delta\times\mathbb{b}_{\text{vix}}[\Delta])}_{\equiv\mathbb{b}_{\text{spread}}[\Delta]}\times\text{V}_t + \underbrace{(\mathbb{c}_{\text{rv}}^{\text{fut}}[\Delta] - \Delta\times\mathbb{c}_{\text{vix}}[\Delta])}_{=0}\times\text{I}_t. \qquad \text{(A43)}
$$

In this case,

$$
\begin{aligned}
\mathbb{a}_{\text{vspread}}[\Delta] &= \{\lambda_{\text{J}}^{\mathbb{P}}(\{\mu_{\text{J}}^{\mathbb{P}}\}^2 \;+\; \{\sigma_{\text{J}}^{\mathbb{P}}\}^2) \;-\; 2\lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1-J_t)\}\times\Delta \\
&\quad +\; (\{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}^2 \;+\; \{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2)\{\frac{\Delta\,\theta_{\text{v}}^{\mathbb{P}}}{\kappa_{\text{v}}^{\mathbb{P}}} + \frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{P}}\,\Delta})}{\kappa_{\text{v}}^{\mathbb{P}}}\{-\frac{\theta_{\text{v}}^{\mathbb{P}}}{\kappa_{\text{v}}^{\mathbb{P}}}\}\} \\
&\quad -\; (\{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}^2 \;+\; \{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2)\{\frac{\Delta\,\theta_{\text{v}}^{\mathbb{Q}}}{\kappa_{\text{v}}^{\mathbb{Q}}} + \frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{Q}}\,\Delta})}{\kappa_{\text{v}}^{\mathbb{Q}}}\{-\frac{\theta_{\text{v}}^{\mathbb{Q}}}{\kappa_{\text{v}}^{\mathbb{Q}}}\}\}, \qquad \text{(A44)}
\end{aligned}
$$

and

$$\mathbb{b}_{\text{vspread}}[\Delta] \;=\; (\{1+\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\,\rho_{\text{v}}\}^2 \;+\; \{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2)\,\{[\frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{P}}\Delta})}{\kappa_{\text{v}}^{\mathbb{P}}}]-[\frac{(1-e^{-\kappa_{\text{v}}^{\mathbb{Q}}\Delta})}{\kappa_{\text{v}}^{\mathbb{Q}}}]\}. \quad \text{(A45)}$$

I have the expressions required for implementation. ■

Characteristic function under $\mathbb{P}$. For $i=\sqrt{-1}$, I can show that

$$\mathbb{H}_t^{\mathbb{P}}[\phi] \;\equiv\; \mathbb{E}_t^{\mathbb{P}}(e^{i\phi\log(\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}})}) \;=\; \exp\Big(\mathbb{a}_{\text{cf}}^{\mathbb{P}}[\phi]+\mathbb{b}_{\text{cf}}^{\mathbb{P}}[\phi]\times \text{V}_t+\mathbb{c}_{\text{cf}}^{\mathbb{P}}[\phi]\times \text{I}_t\Big), \quad \text{(A46)}$$

where

$$\mathbb{a}_{\text{cf}}^{\mathbb{P}}[\phi] \;=\; \mathbb{a}_{\text{J}}[\Delta;\phi]+\mathbb{a}_{\text{v}}[\Delta;\phi]+\mathbb{a}_{\text{I}}[\Delta;\phi], \quad \mathbb{b}_{\text{cf}}^{\mathbb{P}}[\phi] \;=\; \mathbb{b}_{\text{v}}[\Delta;\phi] \quad \text{and} \quad \mathbb{c}_{\text{cf}}^{\mathbb{P}}[\phi] \;=\; \mathbb{b}_{\text{I}}[\Delta;\phi].$$

$$\mathbb{a}_{\text{J}}[\phi] \;=\; -\,i\phi\,\Delta\lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}-1)-\;i\phi\,\Delta\lambda_{\text{J}}^{\mathbb{Q}}\,\mathbb{E}_t^{\mathbb{Q}}(e^{-J_t}-1)+\lambda_{\text{J}}^{\mathbb{Q}}\,\Delta\{\mathbb{E}_t^{\mathbb{Q}}(e^{\{i\phi-1\}J_t})\;-\;1\}, \quad \text{(A47)}$$

The solution is

$$\mathbb{a}_{\text{v}}[\Delta] \;=\; -\frac{\theta_{\text{v}}^{\mathbb{Q}}}{\sigma_{\text{v}}^2}\left[(\psi_{\text{v}}-\{a^*\})\Delta+2\log(1-\frac{(\psi_{\text{v}}-\{a^*\})(1-e^{-\psi_{\text{v}}\Delta})}{2\psi_{\text{v}}})\right]$$

$$\mathbb{b}_{\text{v}}[\Delta] \;=\; \frac{2(A+\frac{1}{2}B^2)(1-e^{-\psi_{\text{v}}\tau})}{2\psi_{\text{v}}-(\psi_{\text{v}}-\{a^*\})(1-e^{-\psi_{\text{v}}\Delta})}$$

$$\alpha^* \;=\; -\;\{-\frac{\chi_{\text{v}}^*}{1}\}+i\phi\,\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\,\rho_{\text{v}}\} \quad \text{(A48)}$$

$$\beta^* \;=\; -\;\{-\frac{\chi_{\text{u}}^*}{1}\}\;+\;i\phi\,\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2} \quad \text{(A49)}$$

$$a^* \;=\; \kappa_{\text{v}}^{\mathbb{Q}}-\sigma_{\text{v}}\,\rho_{\text{v}}\alpha^*-\beta^*\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2} \quad \text{(A50)}$$

$$A \;=\; -\frac{1}{2}\{-\frac{\chi_{\text{v}}^*}{1}\}^2-\frac{1}{2}\{-\frac{\chi_{\text{u}}^*}{1}\}^2 \quad \text{(A51)}$$

$$-\frac{1}{2}i\phi\,\{1+\mathbb{b}_{\text{fut}}[\tau]\sigma_{\text{v}}\,\rho_{\text{v}}\}^2$$

$$-\frac{1}{2}i\phi\,\{\mathbb{b}_{\text{fut}}[\tau]\,\sigma_{\text{v}}\sqrt{1-\rho_{\text{v}}^2}\}^2. \quad \text{(A52)}$$

$$B^2 \;=\; \{\alpha^*\}^2+\{\beta^*\}^2 \quad \text{(A53)}$$

$$\psi_{\text{v}} \;=\; \sqrt{\{a^*\}^2-2\sigma_{\text{v}}^2(A+\frac{1}{2}B^2)}. \quad \text{(A54)}$$

Furthermore,

$$\mathbb{a}_{\mathrm{I}}[\Delta] = -\frac{\theta_{\mathrm{I}}^{\mathbb{Q}}}{\sigma_{\mathrm{I}}^2}\left[(\psi_{\mathrm{I}}-(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B))\Delta+2\log(1-\frac{(\psi_{\mathrm{I}}-(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B))(1-e^{-\psi_{\mathrm{I}}\Delta})}{2\psi_{\mathrm{I}}})\right], \tag{A55}$$

$$\mathbb{b}_{\mathrm{I}}[\Delta] = \frac{2(A+\frac{1}{2}B^2)(1-e^{-\psi_{\mathrm{I}}\Delta})}{2\psi_{\mathrm{I}}-(\psi_{\mathrm{I}}-(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B))(1-e^{-\psi_{\mathrm{I}}\Delta})},$$

$$\psi_{\mathrm{I}} = \sqrt{(\kappa_{\mathrm{I}}^{\mathbb{Q}}-\sigma_{\mathrm{I}}B)^2-2\sigma_{\mathrm{I}}^2(A+\frac{1}{2}B^2)}, \tag{A56}$$

$$A = -\frac{i\phi}{2}\{1+\mathbb{c}_{\mathrm{fut}}[\tau]\,\sigma_{\mathrm{I}}\}^2 \quad \text{and} \quad B = i\phi\{1+\sigma_{\mathrm{I}}\,\mathbb{c}_{\mathrm{fut}}[\tau]\}. \tag{A57}$$

Additionally,

$$\mathbb{E}_t^{\mathbb{Q}}(e^{\{i\phi-1\}J_t}) = \exp(\{i\phi-1\}\mu_{\mathrm{J}}^{\mathbb{Q}}+\frac{1}{2}\{i\phi-1\}^2\{\sigma_{\mathrm{J}}^{\mathbb{Q}}\}^2), \tag{A58}$$

$$\mathbb{E}_t^{\mathbb{Q}}(e^{J_t}) = \exp(\mu_{\mathrm{J}}^{\mathbb{Q}}+\frac{1}{2}\{\sigma_{\mathrm{J}}^{\mathbb{Q}}\}^2) \quad \text{and} \quad \mathbb{E}_t^{\mathbb{Q}}(e^{-J_t}) = \exp(-\mu_{\mathrm{J}}^{\mathbb{Q}}+\frac{1}{2}\{\sigma_{\mathrm{J}}^{\mathbb{Q}}\}^2). \tag{A59}$$

The solution is complete. ■

B. Model of the idiosyncratic component

Let $r_{t+\Delta}^{\mathrm{oil}} = \frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}} - 1 = R_{\{t\to t+\Delta\}}^{\mathrm{oil}} - 1$ and $r_{t+\Delta}^{\mathrm{index}}$ be the return of a commodity futures index. One may contemplate using the S&P GSCI (which is composed of 24 exchange-traded futures contracts on physical commodities). There are no options traded on a commodity index.

Statement (Rationale for positive risk-neutral skewness). Suppose the following holds:

(i) The oil futures return follows a single-index model, where the single index is the commodity futures index;

(ii) The commodity index futures return is essentially symmetric (i.e., $\mathrm{Skewness}_t^{\mathbb{Q},\mathrm{index}} \approx 0$).

Then, risk-neutral oil skewness can switch sign if the idiosyncratic oil skewness switches sign.

Proof. Consider a version of the model adopted in Bakshi, Gao, and Rossi (2019), as follows:

$$\underbrace{r^{\text{oil}}_{t+\Delta}}_{\text{oil futures return}} = \alpha + \beta \underbrace{r^{\text{index}}_{t+\Delta}}_{\substack{\text{return of a}\\ \text{commodity futures index}}} + \underbrace{\varepsilon^{\perp}_{t+\Delta}}_{\substack{\text{idiosyncratic}\\ \text{risk component}}} \quad \text{under } \mathbb{P} \tag{B1}$$

It holds that $\mathbb{E}^{\mathbb{Q}}_t(r^{\text{oil}}_{t+\Delta}) = \mathbb{E}^{\mathbb{Q}}_t(r^{\text{index}}_{t+\Delta}) = 0$ and $\mathbb{E}^{\mathbb{Q}}_t(\varepsilon^{\perp}_{t+\Delta}) = 0$ by construction. Hence,

$$r^{\text{oil}}_{t+\Delta} = \beta\, r^{\text{index}}_{t+\Delta} + \varepsilon^{\perp}_{t+\Delta} \quad \text{under } \mathbb{Q} \tag{B2}$$

The moment generating function satisfies $\mathbb{E}^{\mathbb{Q}}_t(e^{\boldsymbol{\lambda} r^{\text{oil}}_{t+\Delta}}) = \mathbb{E}^{\mathbb{Q}}_t(e^{\boldsymbol{\lambda}\beta r^{\text{index}}_{t+\Delta}}) \times \mathbb{E}^{\mathbb{Q}}_t(e^{\boldsymbol{\lambda}\varepsilon^{\perp}_{t+\Delta}})$. Hence, $\text{Variance}^{\mathbb{Q},\text{oil}}_t = \mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^2) = \beta^2 \text{var}^{\mathbb{Q},\text{index}}_t + \text{var}^{\mathbb{Q},\varepsilon}_t$. Additionally,

$$\begin{aligned}
\text{Skewness}^{\mathbb{Q},\text{oil}}_t &= \frac{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^3)}{\{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^2)\}^{3/2}} && \text{(B3)}\\
&= \frac{\{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t\}^{3/2}\text{Skewness}^{\mathbb{Q},\text{index}}_t + \{\text{var}^{\mathbb{Q},\varepsilon}_t\}^{3/2}\text{Skewness}^{\mathbb{Q},\varepsilon}_t}{\{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t + \text{var}^{\mathbb{Q},\varepsilon}_t\}^{3/2}}. && \text{(B4)}\\
&= \{\mathbb{s}_t\}^{3/2}\text{Skewness}^{\mathbb{Q},\text{index}}_t + \{1-\mathbb{s}_t\}^{3/2}\,\text{Skewness}^{\mathbb{Q},\varepsilon}_t.
\end{aligned}$$

where $\mathbb{s}_t = \beta^2 \text{var}^{\mathbb{Q},\text{index}}_t / \{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t + \text{var}^{\mathbb{Q},\varepsilon}_t\} \in (0,1)$. My result follows if $\text{Skewness}^{\mathbb{Q},\text{index}}_t$ were zero.

The properties of the idiosyncratic risk components affect other risk-neutral central moments. Specifically,

$$\begin{aligned}
\text{Kurtosis}^{\mathbb{Q},\text{oil}}_t - 3 &= \frac{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^4)}{\{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^2)\}^2} - 3 && \text{(B5)}\\
&= \frac{\{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t\}^2\text{Kurtosis}^{\mathbb{Q},\text{index}}_t + \{\text{var}^{\mathbb{Q},\varepsilon}_t\}^2\text{Kurtosis}^{\mathbb{Q},\varepsilon}_t + 6\beta^2\{\text{var}^{\mathbb{Q},\text{index}}_t\}\{\text{var}^{\mathbb{Q},\varepsilon}_t\}}{\{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t + \text{var}^{\mathbb{Q},\varepsilon}_t\}^2} - 3.\\
&= \frac{\{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t\}^2\{\text{Kurtosis}^{\mathbb{Q},\text{index}}_t - 3\} + \{\text{var}^{\mathbb{Q},\varepsilon}_t\}^2\{\text{Kurtosis}^{\mathbb{Q},\varepsilon}_t - 3\}}{\{\beta^2 \text{var}^{\mathbb{Q},\text{index}}_t + \text{var}^{\mathbb{Q},\varepsilon}_t\}^2},\\
&= \{\mathbb{s}_t\}^2\{\text{Kurtosis}^{\mathbb{Q},\text{index}}_t - 3\} + \{1-\mathbb{s}_t\}^2\{\text{Kurtosis}^{\mathbb{Q},\varepsilon}_t - 3\}. && \text{(B6)}
\end{aligned}$$

The presence of idiosyncratic risks contributes to both higher risk-neutral volatility and risk-neutral kurtosis. ■

Table 1

Comparison of model properties across published papers

For brevity of column spacing, I label the models as follows: "✓" indicates the model can capture the indicated feature, "*no*" indicates that the model lacks or omits the feature, and "*na*" implies "not applicable."

- **My estimated model**: There are idiosyncratic, jump, unspanned, and spanned risks. The extracted variables are (i) idiosyncratic variance and (ii) non-idiosyncratic variance.
- Model 1: Gibson and Schwartz (1990) (models of futures prices)
- Model 2: Schwartz (1997) (models of futures prices)
- Model 3: Hilliard and Reis (1998) (model of futures prices)
- Model 4: Casassus and Collin-Dufresne (2005) (models of futures prices)
- Model 5: Kogan, Livdan, and Yaron (2009)
- Model 6: Trolle and Schwartz (2009) (with Heath-Jarrow-Morton setup for cost of carry)
- Model 7: Chiang, Hughen, and Sagi (2015)
- Model 8: Christoffersen, Jacobs, and Li (2016) (discrete-time GARCH framework)
- Model 9: Heath (2019)
- Model 10: Equilibrium models (e.g., Routledge, Seppi, and Spatt (2000), Ready (2018), and Bornstein, Krusell, and Rebelo (2023)).
- Model 11: Crosby and Frau (2022) (with Heath-Jarrow-Morton setup for cost of carry)
- Model 12: Christoffersen, Jacobs, and Pan (2022) (extract state-price density from crude oil option prices)
- Model 13: Gao, Hitzemann, Shaliastovich, and Xu (2022) (macrofinance model with production and consumption)
- Model 14: Jacobs and Li (2023)

	My paper	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Consider idiosyncratic risks	✓	no	no	no	no	no	no	no	no	no	no	no	*na*	no	no
Consider jump risks	✓	no	no	✓	no	no	no	no	✓	no	no	✓	*na*	no	no
Consider unspanned risks	✓	no	no	no	no	no	✓	✓	*na*	no	no	✓	*na*	no	no
Explore model fit to patterns of option risk premiums	✓	no	no	no	no	no	no	no	no	no	no	no	*na*	no	no
Explore model fit to $\{\mathrm{VIX}_t^{\mathrm{oil}}\}^2 = \mathbb{E}_t^{\mathbb{Q}}(\{-\frac{2}{\Delta}\}\log(\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}))$	✓	no	no	no	no	no	no	no	no	no	no	no	no	no	no

Table 2

Excess returns of options on futures on the West Texas Intermediate (WTI) crude oil

I compute the excess returns of OTM puts and calls over 360 monthly option expiration cycles, from January 12, 1990 to December 2019 (the final expiration cycle starts on 12/16/2019 and ends on 01/15/2020). The underlier for the futures is the WTI crude oil and the source of the matched futures and option prices is Barchart.

$$\mathbb{r}^{\text{put}}_{\{t\to t+\Delta\}}[k] = \frac{\max(K - F^{T_F}_{t+\Delta}, 0)}{\text{put}_t[K]} - R^{\text{rf}}_t, \text{ where } K \text{ corresponds to } k = -5\% \text{ and} - 10\%,$$

$$\underbrace{\mathbb{r}^{\text{call}}_{\{t\to t+\Delta\}}[k]}_{\text{monthly (\%)}} = \frac{\max(F^{T_F}_{t+\Delta} - K, 0)}{\text{call}_t[K]} - R^{\text{rf}}_t, \text{ where } K \text{ corresponds to } k = 5\% \text{ and } 10\%.$$

The strike price is K, $F^{T_F}_{t+\Delta}$ is the futures price at option expiration, and R^{rf}_t is the U.S. LIBOR rate (gross rate, monthly). I report the sample mean and 95% bootstrap confidence intervals (using a stationary bootstrap procedure with 10,000 draws). The straddle returns are based on at-the-money puts and calls (i.e., $\mathbb{r}^{\text{straddle}}_{\{t\to t+\Delta\}} = \frac{\max(F^{T_F}_t - F^{T_F}_{t+\Delta}, 0) + \max(F^{T_F}_{t+\Delta} - F^{T_F}_t, 0)}{\text{put}_t[F^{T_F}_t] + \text{call}_t[F^{T_F}_t]} - R^{\text{rf}}_t$). Bootstrap confidence intervals on mean option excess returns that do not bracket zero are indicated by •. Also reported are σ^{Black}, the option implied volatility, and the proportion of option excess returns that are positive "# positive." The reported put (call) delta is $-e^{-r\Delta}\mathcal{N}(-d_1)$ ($e^{-r\Delta}\mathcal{N}(d_1)$), where $d_1 = \frac{1}{\sigma\sqrt{\Delta}}\{-\log(k) + \frac{1}{2}\sigma^2\Delta\}$ (r is the annualized LIBOR rate input and $\Delta = \frac{30}{365}$).

Panel B reports the partitioned average excess returns based on the following regression specification:

$$\mathbb{r}^{\text{call}}_{\{t\to t+\Delta\}}[k] = \begin{cases} \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{low}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{low}}\}} + \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{medium}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{medium}}\}} + \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{high}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{high}}\}} + \underbrace{\varepsilon_{\{t\to t+\Delta\}}}_{\text{error term}}, & \text{or} \\ \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{contango}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{contango}}\}} + \mu^{\text{call}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{backwardation}}\}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{backwardation}}\}} + \varepsilon_{\{t\to t+\Delta\}}. \end{cases}$$

I use proxies for the variable $\mathfrak{s}$, known at the beginning of the option expiration cycles, and the variable construction is described in the text. The p-values rely on the Newey-West HAC estimator (with the lag selected automatically). The oil futures curve is in backwardation for 164 option expiration cycles (44.44%).

	OTM puts		**Straddle**	**OTM calls**	
Moneyness	10%	5%		5%	10%
Option delta	-11.6	-25.1		28.5	13.9
σ^{Black} (%)	37.7	35.8	34.5	34.4	35.5
Panel A: Properties of excess returns of options on oil futures (monthly, %)					
Mean	-22	-8	-1	-6	-36•
Bootstrap Lower	-53	-31	-9	-30	-66
Bootstrap Upper	6	13	7	15	-11
(NW[P])	(0.164)	(0.484)	(0.844)	(0.596)	(0.026)
# positive (out of 360)	45	77	155	87	37
Panel B: Properties of conditional excess returns of options on oil futures (monthly, %)					
$\text{VIX}^{\text{SP\&500}}_t$ is *Low*	-27	-6	1	-13	-44
$\text{VIX}^{\text{SP\&500}}_t$ is *Medium*	-7	6	6	5	-32
$\text{VIX}^{\text{SP\&500}}_t$ is *High*	-32	-24	-10•	-11	-33•
$\text{VIX}^{\text{oil}}_t$ is *Low*	5	24	13•	-4	-48
$\text{VIX}^{\text{oil}}_t$ is *Medium*	-22	-16	1	8	-14
$\text{VIX}^{\text{oil}}_t$ is *High*	-48•	-33•	-16•	-24•	-47•
Oil futures curve is in *backwardation*	-61•	-36•	-2	19	-11
Oil futures curve is in *contango*	11	15	0	-27•	-57•

Table 3

Evidence from a matched sample of options on natural gas, heating oil, and WTI crude oil

This table presents results on option excess returns for a matched sample, where the futures underliers are natural gas, heating oil, and WTI crude oil. The data is from OptionMetrics and the sample period is from January 2005 to December 2020, with 191 monthly option expiration cycles. I compute the excess returns of OTM puts and calls over option expiration cycles, as follows:

$$\mathbb{r}^{\text{put}}_{\{t\to t+\Delta\}}[k] = \frac{\max(K - F^{T_F}_{t+\Delta}, 0)}{\text{put}_t[K]} - R^{\text{rf}}_t, \text{ where } K \text{ corresponds to } k = -5\% \text{ and} - 10\%,$$

$$\underbrace{\mathbb{r}^{\text{call}}_{\{t\to t+\Delta\}}[k]}_{\text{monthly (\%)}} = \frac{\max(F^{T_F}_{t+\Delta} - K, 0)}{\text{call}_t[K]} - R^{\text{rf}}_t, \text{ where } K \text{ corresponds to } k = 5\% \text{ and } 10\%.$$

The strike price is K, $F^{T_F}_{t+\Delta}$ is the futures price at option expiration, and R^{rf}_t is the U.S. LIBOR rate (gross rate, monthly). The straddle returns are based on at-the-money puts and calls. I report the sample mean and 95% bootstrap confidence intervals (using a stationary bootstrap procedure with 10,000 draws). Bootstrap confidence intervals on mean option excess returns that do *not* bracket zero are indicated by •. Also reported are σ^{Black}, the option implied volatility. The reported put (call) delta is $-e^{-r\Delta}\mathcal{N}(-d_1)$ ($e^{-r\Delta}\mathcal{N}(d_1)$), where $d_1 = \frac{1}{\sigma\sqrt{\Delta}}\{-\log(k) + \frac{1}{2}\sigma^2\Delta\}$ (r is the annualized LIBOR rate input and $\Delta = \frac{30}{365}$).

Panel A: Properties of excess returns of options on natural gas (Henry Hub, monthly (%))

	OTM puts		Straddle	OTM calls	
Moneyness	10%	5%		5%	10%
Option delta	-17.9	-33.8		38.5	21.8
σ^{Black} (%)	47.0	46.9	46.9	47.5	48.4
Mean	5	5	-9	$\boxed{-32}$	$\boxed{-44}$
Bootstrap Lower	-33	-22	-19	-56	-73
Bootstrap Upper	39	30	1	-9	-18

Panel B: Properties of excess returns of options on heating oil (NY Harbor ULSD, monthly (%))

	OTM puts		Straddle	OTM calls	
Moneyness	10%	5%		5%	10%
Option delta	-10.4	-23.9		26.8	12.2
σ^{Black} (%)	34.4	33.1	32.1	32.1	32.7
Mean	3	-1	-1	$\boxed{-22}$	$\boxed{-56}$
Bootstrap Lower	-60	-38	-12	-46	-82
Bootstrap Upper	56	32	9	-1	-35

Panel C: Properties of excess returns of options on WTI crude oil (monthly (%))

	OTM puts		Straddle	OTM calls	
Moneyness	10%	5%		5%	10%
Option delta	-11.5	-25.0		28.1	13.7
σ^{Black} (%)	39.1	36.8	35.3	34.6	35.2
Mean	19	16	1	$\boxed{-24}$	$\boxed{-64}$
Bootstrap Lower	-48	-25	-11	-50	-86
Bootstrap Upper	75	52	12	-2	-45

Table 4

Risk-neutral return moments extracted from options on futures on the WTI crude oil

The underlier for the futures is the WTI crude oil and the source of the matched futures and option prices is Barchart. Let $R^{\text{oil}}_{\{t\to t+\Delta\}} \equiv \frac{F^{T_F}_{t+\Delta}}{F^{T_F}_t}$ and hence $\mathbb{E}^{\mathbb{Q}}_t(R^{\text{oil}}_{\{t\to t+\Delta\}}) = 1$. I perform the following calculations using option prices (the futures is the underlier):

$$\text{Variance}^{\mathbb{Q}}_t = \mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^2), \quad \text{Skewness}^{\mathbb{Q}}_t = \frac{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^3)}{\{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^2)\}^{3/2}} \quad \text{and} \quad \text{Kurtosis}^{\mathbb{Q}}_t = \frac{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^4)}{\{\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^2)\}^2},$$

where it holds, for $n = 2,3,4$, that

$$\mathbb{E}^{\mathbb{Q}}_t(\{R^{\text{oil}}_{\{t\to t+\Delta\}} - 1\}^n) = \int_{K<F^{T_F}_t} \mathbb{w}[K]\,\text{put}_t[K]dK + \int_{K>F^{T_F}_t} \mathbb{w}[K]\,\text{call}_t[K]\,dK, \quad \text{where } \mathbb{w}[K] = \frac{(n-1)n(\frac{K}{F^{T_F}_t} - 1)^{n-2}}{(F^{T_F}_t)^2}.$$

The sample period is from January 12, 1990 to December 2019, with 360 monthly option expiration cycles. The final expiration cycle starts on 2019/12/16 and ends on 2020/01/15. The average days to expiration (i.e., $\Delta = \frac{30}{365}$) is 30 days. Both $\text{VIX}^{\text{oil}}_t$ and risk-neutral oil volatility (i.e., $\sqrt{\text{Variance}^{\mathbb{Q}}_t}$) are in annual percentage units. I compute $\{\text{VIX}^{\text{oil}}_t\}^2 = \frac{2R^{\text{rf}}_t}{\frac{30}{365}} \int_{K<F^{T_F}_t} \frac{\text{put}_t[K]}{K^2}\,dK + \frac{2R^{\text{rf}}_t}{\frac{30}{365}} \int_{K>F^{T_F}_t} \frac{\text{call}_t[K]}{K^2}\,dK$. ACF_1 is the first-order autocorrelation and SD is the standard deviation.

	Mean	Bootstrap Lower	Bootstrap Upper	SD	Min.	Percentiles 5th	25th	50th	75th	95th	Max.	ACF_1
$\text{VIX}^{\text{oil}}_t$ (annualized)	**36.9**	35.6	38.2	12.97	17.0	20.5	27.5	35.1	42.6	58.0	109.8	0.81
$\sqrt{\text{Variance}^{\mathbb{Q}}_t}$ (annualized)	**37.15**	35.7	38.5	13.1	16.3	21.1	28.3	35.3	43.3	59.5	107.6	0.82
$\text{Skewness}^{\mathbb{Q}}_t$ (monthly)	**0.38**	0.33	0.42	0.43	-1.7	-0.3	0.2	0.4	0.6	1.1	2.3	0.38
$\text{Kurtosis}^{\mathbb{Q}}_t$ (monthly)	**6.34**	5.9	6.7	4.09	1.4	2.4	3.5	5.1	7.8	13.4	27.8	0.73

Table 5

Summary picture of model fit and actual values

The sample period from January 12, 1990 to December 2019, with 360 monthly option expiration cycles, is used to compute the model determined quantities at the beginning of each oil option expiration cycle with extracted V_t and I_t, and estimated parameter vector Θ (as per Table 6). The final expiration cycle starts (ends) on 2019/12/16 (2020/01/15). Summary statistics such as mean, standard deviation (SD), and the percentiles are reported.

						Percentiles					
			Mean	SD	Min.	5th	25th	50th	75th	95th	Max.
1	$\mathrm{VIX}_t^{\mathrm{oil}}$	Model	37.4	13.2	16.3	21.2	28.5	35.6	43.7	60.0	106.9
		Actual	36.9	13.0	17.0	20.5	27.5	35.1	42.6	58.0	109.8
2	$\sqrt{\mathrm{Variance}_t^{\mathbb{Q}}}$	Model	37.1	13.1	16.3	21.1	28.3	35.3	43.3	59.5	107.6
		Actual	37.1	13.1	16.3	21.1	28.3	35.3	43.3	59.5	107.6
3	$100 \times \frac{\text{10\% OTM put price}}{\text{futures price}}$	Model	1.1	1.1	0.0	0.2	0.5	0.8	1.3	2.7	9.9
		Actual	1.2	1.0	0.0	0.2	0.5	1.0	1.5	3.0	7.7
4	$100 \times \frac{\text{5\% OTM put price}}{\text{futures price}}$	Model	2.1	1.4	0.2	0.6	1.2	1.8	2.5	4.3	11.8
		Actual	2.2	1.3	0.2	0.6	1.3	2.0	2.8	4.4	9.4
5	$100 \times \frac{\text{ATM put price}}{\text{futures price}}$	Model	3.9	1.6	1.3	2.0	3.0	3.7	4.5	6.4	13.8
		Actual	4.2	1.5	1.1	2.2	3.1	4.0	4.9	6.7	11.3
6	$100 \times \frac{\text{ATM call price}}{\text{futures price}}$	Model	4.0	1.6	1.2	1.9	3.1	3.7	4.5	6.7	14.7
		Actual	4.2	1.6	1.4	2.2	3.2	4.0	4.9	6.9	12.2
7	$100 \times \frac{\text{5\% OTM call price}}{\text{futures price}}$	Model	2.1	1.5	0.2	0.5	1.2	1.8	2.6	4.5	12.5
		Actual	2.2	1.4	0.3	0.6	1.2	2.0	2.8	4.7	9.1
8	$100 \times \frac{\text{10\% OTM call price}}{\text{futures price}}$	Model	1.1	1.2	0.0	0.1	0.4	0.7	1.3	3.0	11.6
		Actual	1.1	1.1	0.0	0.2	0.4	0.8	1.4	2.9	7.9
9	Futures return volatility	Model	35.4	12.3	15.7	20.2	27.1	33.6	41.3	56.4	100.3
		Actual	32.2	15.1	9.8	15.0	22.7	29.2	37.7	61.4	133.7
10	Spot return volatility	Model	35.4	12.3	15.7	20.2	27.1	33.7	41.3	56.5	100.6
		Actual	34.9	18.4	11.7	15.9	24.1	30.9	39.6	71.4	176.4

Table 6

Model estimates, properties of extracted idiosyncratic volatility, and economic implications

This table presents parameter estimates and asymptotic standard errors (in parentheses) for models estimated using maximum likelihood and Kalman filter over option expiration cycles. Five measurements are taken into consideration (as shown in equation (29)): (i) $\log(\frac{F_t^{T_F}}{S_t})$, where the futures price with a constant maturity of 30-days, (ii) $\log(1+\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}-1\}^2))$, where $\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}-1\}^2$ is the 30-day risk-neutral futures variance, (iii) 30-day expected integrated variance of futures returns (with a maturity of 30 days), (iv) 30-day expected integrated variance of spot returns, and (v) variance spreads. The maximized log-likelihood values and the summary statistics for the extracted $\sqrt{V_t}$ and $\sqrt{I_t}$ (annualized, %) are shown.

Panel A: Features of extracted idiosyncratic variance and non-idiosyncratic variance

	Mean	SD	Min.	Percentiles 5th	25th	50th	75th	95th	Max.
$\sqrt{V_t}$ (annualized, %)	33	14	5	14	23	31	40	57	103
$\sqrt{I_t}$ (annualized, %)	11	3	4	7	9	10	12	17	28

	Panel B: Idiosyncratic variance			**Panel C: Jump component**					
	$\kappa_I^{\mathbb{P}}$	$\theta_I^{\mathbb{P}}$	σ_I	$\mu_J^{\mathbb{P}}$	$\sigma_J^{\mathbb{P}}$	$\lambda_J^{\mathbb{P}}$	μ_J^{rp}	σ_J^{rp}	λ_J^{rp}
Estimate	7.049	0.083	0.855	0.080	0.003	0.859	0.008	-0.004	0.131
(Standard Error)	(1.604)	(0.051)	(0.016)	(0.021)	(0.017)	(0.434)	(0.111)	(5.412)	(0.249)

	Panel D: Non-idiosyncratic variance			**Panel E: Risk premium components of variance**		
	$\kappa_v^{\mathbb{P}}$	$\theta_v^{\mathbb{P}}$	σ_v	χ_v^*	χ_u^*	ρ_v
Estimate	3.432	0.339	0.722	3.337	-3.304	-0.677
(Standard Error)	(0.000)	(0.033)	(0.021)	(0.947)	(1.006)	(0.000)

	Panel F: Cost of carry			**Panel G: Measurement error**					
	δ_v°	δ_I^*	δ_v^*	$\sqrt{\Omega_1}$	$\sqrt{\Omega_2}$	$\sqrt{\Omega_3}$	$\sqrt{\Omega_4}$	$\sqrt{\Omega_5}$	$\mathbb{L}[\Theta]$
Estimate	0.010	0.049	0.074	0.021	0.000	0.008	0.014	0.008	5836
(Standard Error)	(0.017)	(0.525)	(0.080)	(0.001)	(0.000)	(0.000)	(0.001)	(0.000)	

Table 7

Option excess returns and principal components of futures returns and extracted idiosyncratic variance

Reported are the results from the following regressions (and likewise for OTM puts and calls):

$$\underbrace{\mathbb{r}^{\text{straddle}}_{\{t\to t+\Delta\}}}_{\text{excess returns of straddle on WTI}} = \text{constant} + \sum_{\ell=1}^{3} \Lambda_{\text{pc}}^{[\ell],\text{return}} \times \underbrace{\widehat{\text{PC}}^{[\ell],\text{return}}_{\{t\to t+\Delta\}}}_{\text{PC of oil futures returns}} + \Upsilon_{\text{idio}} \times \underbrace{\{\text{I}_{t+\Delta} - \text{I}_t\}}_{\text{model extracted}} + \underbrace{\varepsilon_{\{t\to t+\Delta\}}}_{\text{residual}},$$

where $\widehat{\text{PC}}^{[\ell],\text{return}}_{\{t\to t+\Delta\}}$, for $\ell = 1,\ldots,3$, are the principal components extracted from the returns of oil futures over option expiration cycles. The p-values rely on the Newey-West HAC estimator (with the lag selected automatically).

	Panel A: Spanning regressions augmented with $\{\text{I}_{t+\Delta} - \text{I}_t\}$				
	OTM Puts		Straddle	OTM Calls	
	10	5		5	10
$\Lambda_{\text{pc}}^{[1],\text{return}}$	-8.0	-7.6	0.16	7.5	7.0
(NW[p])	(0.00)	(0.00)	(0.62)	(0.00)	(0.00)
$\Lambda_{\text{pc}}^{[2],\text{return}}$	1.57	3.94	-4.13	-19.4	-24.7
(NW[p])	(0.70)	(0.15)	(0.01)	(0.01)	(0.03)
$\Lambda_{\text{pc}}^{[3],\text{return}}$	11.8	5.7	-0.64	-21.2	-32.4
(NW[p])	(0.31)	(0.51)	(0.89)	(0.36)	(0.37)
Υ_{idio}	36.0	27.6	11.9	16.8	20.2
(NW[p])	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)
constant$\times$100	-21	-7	0	-6	-36
(NW[p])	(0.08)	(0.30)	(0.89)	(0.47)	(0.00)
$\overline{R}^2$ (%)	36	52	21	48	34
$\overline{R}^2$ (restricted $\Upsilon_{\text{idio}} = 0$, %) (when risks are spanned by oil futures)	27	43	6	45	31

Panel B: Regression is $\text{Actual}_t = \Pi_0 + \Pi_{\text{I}} \log(\text{I}_t) + \Pi_{\text{V}} \log(\text{V}_t) + \widetilde{e}_t$

Dependent Variable (from data)		$\log(\text{I}_t)$ Coef.	$\log(\text{I}_t)$ NW[p]	$\log(\text{V}_t)$ Coef.	$\log(\text{V}_t)$ NW[p]	Π_0 Coef.	Π_0 NW[p]	$\overline{R}^2$ (%)	CORR with $\log(\text{I}_t)$
$\log(\text{VIX}_t^{\text{oil}})$	Univariate	1.06	(0.00)			1.34	(0.00)	81	0.90
	Bivariate	0.10	(0.10)	0.64	(0.00)	-0.04	(0.70)	95	
$\log(\text{Variance}_t^{\mathbb{Q}})$	Univariate	1.08	(0.00)			5.99	(0.00)	83	0.91
	Bivariate	0.10	(0.04)	0.65	(0.00)	4.58	(0.00)	97	

Table 8

Information content of the option pricing model for the option prices differentiated by moneyness

At the beginning of each oil option expiration cycle, equations (37) and (38) are used to compute the put and call prices, respectively, with extracted V_t and I_t and estimated parameter vector Θ (as per Table 6). The p-value on coefficient — shown in parentheses — is calculated using the Newey-West HAC estimator (with the lag selected automatically). The sample period is from January 1990 to December 2020 with 360 monthly option expiration cycles. In the regression analysis, $PC_t^{[j],\text{return}}$ ($PC_t^{[j],\text{variance}}$) correspond to three (two) principal components of crude oil futures returns (return variances). The return variance principal components are extracted from five monthly variances (using daily futures returns with maturities of 1-, 3-, 6-, 9-, and 12-month).

	Panel A: Dollar Option Pricing Errors (model - actual)			**Panel B Percentage Option Pricing Errors** ($100 \times \log(\frac{\text{model}}{\text{actual}})$)			**Panel C: Regression** $\log(\frac{\text{actual}_t}{\text{futures}_t}) = \alpha + \beta \log(\frac{\text{model}_t}{\text{futures}_t}) + \widetilde{e}_t$				
	Average	Absolute	RMSE	Average	Absolute	RMSE	α	NW[p]	β	NW[p]	R^2
10% OTM puts	\$0.09	\$0.13	\$0.25	9	27	36	-0.05	0.83	0.97	0.00	83
5% OTM puts	\$0.15	\$0.19	\$0.34	10	19	25	-0.15	0.40	0.94	0.00	86
ATM puts	\$0.18	\$0.22	\$0.38	5	11	13	-0.06	0.60	0.96	0.00	88
ATM calls	\$0.06	\$0.11	\$0.22	6	10	13	-0.17	0.06	0.93	0.00	89
5% OTM calls	\$0.13	\$0.18	\$0.32	8	19	25	-0.34	0.02	0.89	0.00	87
10% OTM calls	\$0.18	\$0.22	\$0.38	2	27	36	-0.32	0.16	0.93	0.00	85

Panel D: Cross-sectional fit to option prices across moneyness

$100 \times \log(\frac{\text{model}}{\text{actual}})$	Mean	SD	Min	5	25	50	75	95	Max.
Average errors (all options)	7	21	-61	-33	-2	6	20	40	65
Absolute errors (all options)	19	14	0	3	8	17	27	43	65
RMSE (all options)	22	15	0	4	9	19	30	48	78

Panel E: $\log(\frac{\text{model}}{\text{actual}}) = \text{constant} + \sum_{j=1}^{3} \mathbb{b}_{\text{pc}}^{[j],\text{return}} \times PC_{t-1}^{[j],\text{return}} + \sum_{j=1}^{2} \mathbb{b}_{\text{pc}}^{[j],\text{variance}} \times PC_{t-1}^{[j],\text{variance}} + \widetilde{e}_t$

	10% OTM put	5% OTM put	ATM put	ATM call	5% OTM call	10% OTM call
constant$\times$100	9	10	5	6	8	2
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.45)
$\overline{R}^2$ (%)	3	4	5	6	5	3

Table 9

Information content of the model for risk-neutral return moments and futures

At the beginning of each oil option expiration cycle, the model determined quantities are computed with extracted V_t and I_t and estimated parameter vector Θ (as per Table 6). The sample period is from January 1990 to December 2020 (360 monthly option expiration cycles). The p-value on coefficient — shown in parentheses — is calculated using the Newey-West HAC estimator (with the lag selected automatically). In the regression analysis, $\mathrm{PC}_t^{[j],\mathrm{return}}$ ($\mathrm{PC}_t^{[j],\mathrm{variance}}$) correspond to three (two) principal components of crude oil futures returns (return variances). The return variance principal components are extracted from five monthly variances (using daily futures returns with maturities of 1-, 3-, 6-, 9-, and 12-month).

Panel A: Model performance and dynamic fit

	model - actual			$\log(\mathrm{actual}_t) = \alpha + \beta \log(\mathrm{model}_t) + \widetilde{e}_t$				
	Average	Absolute	RMSE	α	NW[p]	β	NW[p]	$\overline{R}^2$ (%)
$\mathrm{VIX}_t^{\mathrm{oil}}$	0.5	1.3	1.7	-0.04	0.00	0.98	0.00	98
$\sqrt{\mathrm{Variance}_t^{\mathbb{Q}}}$	0.0	0.0	0.0	0.4	0.00	1.0	0.00	100
Futures return volatility	3.1	7.7	10.0	-0.04	0.85	0.97	0.00	58
Spot return volatility	0.5	8.9	12.4	0.12	0.59	0.95	0.00	51
Log of futures price	0.0	0.0	0.0	-0.00	0.77	1.0	0.00	100

Panel B: $\mathrm{Model}_t - \mathrm{Actual}_t = \mathrm{constant} + \sum_{j=1}^{3} \mathbb{b}_{\mathrm{pc}}^{[j],\mathrm{return}} \times \mathrm{PC}_{t-1}^{[j],\mathrm{return}} + \sum_{j=1}^{2} \mathbb{b}_{\mathrm{pc}}^{[j],\mathrm{variance}} \times \mathrm{PC}_{t-1}^{[j],\mathrm{variance}} + \widetilde{e}_t$

	$\mathrm{VIX}_t^{\mathrm{oil}}$	$\sqrt{\mathrm{Variance}_t^{\mathbb{Q}}}$	Futures Volatility	Spot Volatility	$\log(\mathrm{Futures}_t)$
constant	0.5	0.0	3.1	0.6	0.0
	(0.000)	(0.05)	(0.00)	(0.46)	(0.27)
$\overline{R}^2$ (%)	0.7	2	4	4	19

Table 10

Option risk premiums from the model

This table presents the *mean* option risk premiums implied by the model over the sample period of January 12, 1990 to December 2019, with 360 option expiration cycles. At the start of each cycle, the model parameters and V_t and I_t are used to compute the option risk premiums, as follows (for $\mathbb{x} \equiv \log(\frac{F^{T_F}_{t+\Delta}}{F^{T_F}_{t-}})$ and $k = \log(\frac{K}{F^{T_F}_t})$):

$$\begin{aligned}
\text{Risk Premium}^{\text{put}}_{\{t\to t+\Delta\}} &\equiv R^{\text{rf}}_{\{t\to t+\Delta\}}\Big(\frac{\mathbb{E}^{\mathbb{P}}_t(\max(e^k - e^{\mathbb{x}},0))}{\mathbb{E}^{\mathbb{Q}}_t(\max(e^k - e^{\mathbb{x}},0))} - 1\Big),\\
\text{Risk Premium}^{\text{straddle}}_{\{t\to t+\Delta\}} &\equiv R^{\text{rf}}_{\{t\to t+\Delta\}}\Big(\frac{\mathbb{E}^{\mathbb{P}}_t(\max(1 - e^{\mathbb{x}},0)) + \mathbb{E}^{\mathbb{P}}_t(\max(e^{\mathbb{x}} - 1,0))}{\mathbb{E}^{\mathbb{Q}}_t(\max(1 - e^{\mathbb{x}},0)) + \mathbb{E}^{\mathbb{Q}}_t(\max(e^{\mathbb{x}} - 1,0))} - 1\Big),\\
\text{Risk Premium}^{\text{call}}_{\{t\to t+\Delta\}} &\equiv R^{\text{rf}}_{\{t\to t+\Delta\}}\Big(\frac{\mathbb{E}^{\mathbb{P}}_t(\max(e^{\mathbb{x}} - e^k,0))}{\mathbb{E}^{\mathbb{Q}}_t(\max(e^{\mathbb{x}} - e^k,0))} - 1\Big).
\end{aligned}$$

The expressions for $\mathbb{E}^{\mathbb{P}}_t(\max(e^{\mathbb{x}} - e^k,0))$ and $\mathbb{E}^{\mathbb{Q}}_t(\max(e^{\mathbb{x}} - e^k,0))$ are provided in (43)–(44). Reported are the mean option risk premiums.

	10 OTM put (%)	Straddle (%)	10% OTM call (%)
Lower Bootstrap (data)	-53	-9	-66
Upper Bootstrap (data)	6	7	-11
Option Risk Premium (model) (mean)	$\boxed{-16}$	$\boxed{-1}$	$\boxed{-14}$

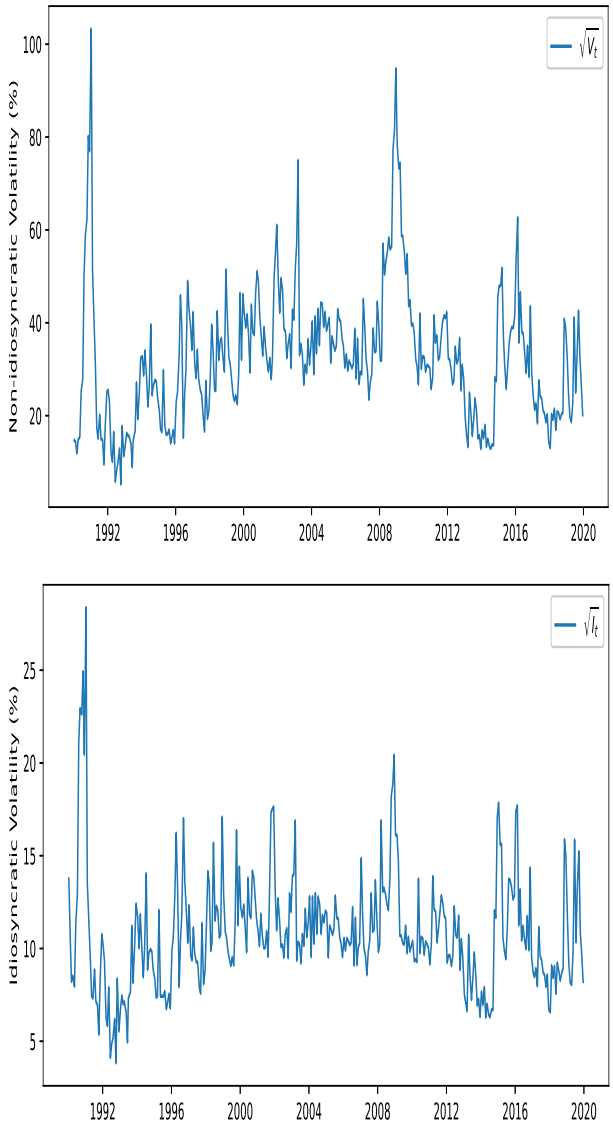

Figure 1. **Filtered non-idiosyncratic volatility ($\sqrt{V_t}$) and idiosyncratic volatility ($\sqrt{I_t}$)**

Using Kalman filter and maximum likelihood, the five measurements (as shown in equation (28)) are taken into consideration: (i) $\log(\frac{F_t^{T_F}}{S_t})$, where the futures price with a constant maturity of 30-days, (ii) $\log(1+\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}-1\}^2))$, where $\mathbb{E}_t^{\mathbb{Q}}(\{\frac{F_{t+\Delta}^{T_F}}{F_t^{T_F}}-1\}^2$ is the 30-day risk-neutral futures variance, (iii) 30-day expected integrated variance of futures returns (with a maturity of 30 days), (iv) 30-day expected integrated variance of spot returns, and (v) variance spreads. The estimations are done over option expiration cycles.

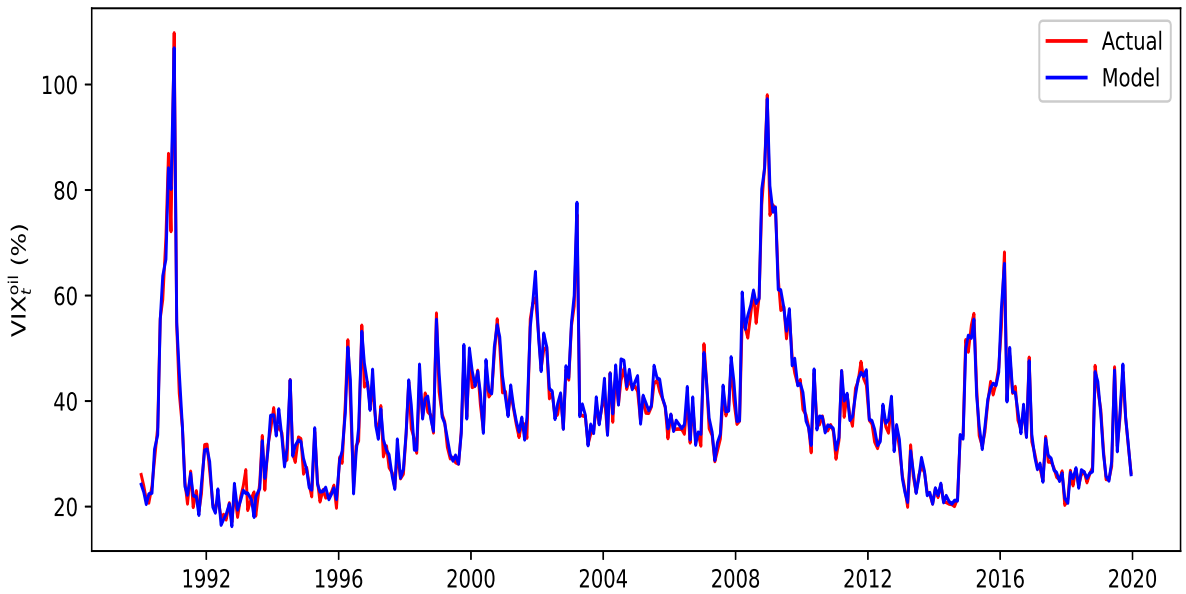

Figure 2. **Model-based** $\mathrm{VIX}_t^{\mathbf{oil}}$

I utilized 360 option expiration cycles of data from January 1990 to December 2019 to estimate the model (see Table 6). Plotted are the model-based and data-based $\mathrm{VIX}_t^{\mathrm{oil}}$ values. The model-based $\mathrm{VIX}_t^{\mathrm{oil}}$ is calculated (following Result 2) as

$$\textbf{Model}: \quad \{\mathrm{VIX}_t^{\mathrm{oil}}\}^2 = \mathbb{a}_{\mathrm{vix}}[\Delta] \; + \; \mathbb{b}_{\mathrm{vix}}[\Delta] \times \mathrm{V}_t \; + \; \mathbb{c}_{\mathrm{vix}}[\Delta] \times \mathrm{I}_t, \qquad \text{for } \; \Delta = \frac{30}{365},$$

where $\mathbb{a}_{\mathrm{vix}}[\Delta]$, $\mathbb{b}_{\mathrm{vix}}[\Delta]$, and $\mathbb{c}_{\mathrm{vix}}[\Delta]$ are presented in (A13)–(A15) of Appendix A. The data-based $\mathrm{VIX}_t^{\mathrm{oil}}$ — the options mimicking portfolio for the log contract — is obtained via the options portfolio as

$$\textbf{Data}: \quad \{\mathrm{VIX}_t^{\mathrm{oil}}\}^2 = \frac{2R_t^{\mathrm{rf}}}{\frac{30}{365}} \int_{K<F_t^{T_F}} \frac{\mathrm{put}_t[K]}{K^2}\, dK + \frac{2R_t^{\mathrm{rf}}}{\frac{30}{365}} \int_{K>F_t^{T_F}} \frac{\mathrm{call}_t[K]}{K^2}\, dK.$$

The plotted values of $\mathrm{VIX}_t^{\mathrm{oil}}$ are in units of annualized percentage.